CCfits Reference Manual

1.1.1

Generated by Doxygen 1.2.7

Thu Sep 19 15:34:43 2002

Contents

1 CCfits Documentation 1

2 CCfits User's Guide 3

3 CCfits Module Index 31

4 CCfits Hierarchical Index 31

5 CCfits Compound Index 33

6 CCfits Module Documentation 35

7 CCfits Namespace Documentation 37

8 CCfits Class Documentation 37

1 CCfits Documentation

1.1 Introduction

CCfits is an object oriented interface to the cfitsio library. cfitsio is a widely used library for manipulating FITS (Flexible Image Transport System) formatted files. This following documentation assumes prior knowledge of the FITS format and some knowledge of the use of the cfitsio library, which is in wide use, well developed, and available on many platforms.

Readers unfamiliar with FITS but in need of performing I/O with FITS data sets are directed to the first cfitsio manual, available at *http://heasarc.gsfc.nasa.gov/docs/software/fitsio/fitsio.html* Information about the FITS file format and the current standard is available from *http://fits.gsfc.nasa.gov*

The CCfits library provides an interface that allows the user to manipulate FITS format data through the high-level building blocks of FITS files and Header-Data Units (HDUs). The implementation is designed to hide the details of performing FITS I/O from the user, who will write calls that manipulate FITS objects by passing filenames and lists of strings that represent HDUs, keywords, image data and data columns. Unlike cfitsio, which typically requires several calls to access data *(e.g. open file, move to correct header, determine column containing table data, read data)* CCfits is designed to make reading data atomic. For example, it exploits internally existing optimization techniques for FITS I/O, choosing the optimal reading strategy as available [see the

cfitsio manual, Chapter 13] when data are read on initialization. Data written by CCfits will also be compliant with the FITS standard by specification of class constructors representing FITS dataset elements.

CCfits necessarily works in a fundamentally different way than cfitsio. The general pattern of usage for CCfits is: create a FITS object, which either opens a disk file or creates a new disk file, create references to existing or new HDU objects within it, and manipulated the data through the references. For files with Write access the library is designed to keep the FITS object on disk in sync with the memory copy. The additional memory copy increases the resources required by a calling program in return for some flexibility in accessing the data.

1.2 About this Manual

This document lays out the specification for the CCfits library.

The next sections document the installation procedure and the demonstration program *cookbook* which gives examples of usage with comments.

- Installing the Package
- Getting Started
- Implementation Notes
- What's Present, What's Missing

Following sections give a list of what is implemented in CCfits compared to the cfitsio library. For background information and as an example there is a section describing how CCfits is to be used in XSPEC, for which it was originally designed, which may serve to give the reader some insight into the design decisions made.

1.3 Authors and Acknowledgements

CCfits was written as part of a re-engineering effort for the X-Ray data analysis program, XSPEC. It was designed using Rational Rose and originally implemented on a Solaris platform by Ben Dorman (ben.dorman@gsfc.nasa.gov) to whom blame should be attached. Sandhia Bansal worked on part of the implementation, and Paul Kunz (pfkeb@slac.stanford.edu) wrote the configuration scheme and dispensed helpful advice. Thanks to R. Mathar (MPIA) and P. Jonsson (Lick Obs.) for contributing many helpful suggestions, and ports to HP-UX and AIX respectively.

Suggestions and bug reports are welcome, as are offers to fill out parts of the implementation that are missing. We are also interested in knowing which parts of cfitsio that are not currently supported should be the highest priority for future extensions.

2 CCfits User's Guide

2.1 User Guide Contents

Installing the Package 3

Implementation Notes 7

Xspec and CCfits 9

Getting Started 10

Writing The Primary Image 12

Creating and Writing to an Ascii Table Extension 15

Creating and Writing to a Binary Table Extension 19

Copying an Extension between Files 24

Selecting Table Data 25

Reading Header information from a HDU 26

Reading an Image 27

Reading a Table Extension 28

What's Present, What's Missing 29

Todo List 30

2.2 Installing the Package

2.2.1 Platforms

CCfits has been developed and tested primarily on Solaris using the SunPro CC compiler 5.1 [Sun Workshop/Forte 6.0] *prior versions of SunPro CC will not compile CCfits as they do not support templated member functions.*

CCfits has also been built and tested on Linux on the RedHat 6.2 and 7.1 distributions using g++ 2.95.3 and 3.0, and on OSF1 v. 4.0D with g++ /2.95.3. CCfits will also build with gcc 2.95.2.

In summary, this release has been tested on the following platforms:

Platform	Compiler	Optimization
Solaris 7	SunPro CC 5.1, 5.3	-g and -fast
OSF1/4.0D	g++ /2.95.3	-g -O2
Linux/RedHat 7.1,6.2	g++ /2.95.3, 3.0.x, 3.1, 3.2	-g -O2
HP-UX 10.20	g++ /2.95.x	-g -O2
AIX	kcc	(use Makefile.kcc)

N.B. For exception support, the g++ compiler needs to be built with threads enabled (–enable-threads or similar option). While this is supposed to be the default on some platforms, it is wise to ensure that it is turned on by explicitly specifying it when the compiler is (re)-built. If this is not done, CCfits may dump core whenever the first exception is thrown in your code.

Not yet supported are: Solaris with gcc compilers (builds, standand library related run time error in test program); Mac OS X / Darwin (compiler distributed with OS X does not have valarray header: not yet tested on gcc 3.1); Windows 2000/ VC++.net (library build successfully; standard library related runtime error in test program).

To build and install CCfits from source code on a UNIX-like (e.g. UNIX, Linux, or Cygwin) platform, take the following steps.

2.2.2 Autogen

If you obtained the source from the CVS repository, you need to do this first step. If you obtained the source from a gzipped tar file, skip to the step 2.

If you want to do a build with a compiler other than GCC, then you need to turn off automatic dependency generation which only works with GCC. Edit the Makefile.am file by adding 'no-dependencies' to the line

AUTOMAKE_OPTIONS = foreign

If you do not do this, some options will be passed to compiler that it might not understand. At least with the Sun C++ compiler, these options are ignored but lead to a warning message for each file that is compiled. Other compilers may treat it as an error.

Type in the following in the toplevel source directory:

> ./autogen

autogen is a UNIX shell script that runs a series of commands in a particular order. It requires automake, autoconf and libtool to be installed on your system.

Note that you will obtain the following message after running 'autogen':

configure.in: 36: required file 'config/ltconfig' not found

Disregard this message. It is not an error.

2.2.3 Configure

By default, the g++ compiler and linker will be used. If you want to compile and link with a different compiler and linker, you can set some environment variable before running the configure script. For example, to use Sun's C++ compiler, do the following if you are a csh user:

> setenv CXX CC

and the equivalent if you are not. You can set the absolute path to the compiler you want to use if necessary.

By default, the configuration sets the compiler optimization flags environment variable to -g -O2 for g++ and -g for other compilers. If you wish to override this (for example, to use -O4 or -fast for SunPro CC or to switch off the optimization on g++), set this variable before running configure.

CCfits requires that the 'cfitsio' package, version 2.4 or later, is available on your system. The configure script that you will run takes an option to specify the location of the cfitsio package. The current version has been tested with cfitsio v2.410.

If the 'cfitsio' package is installed in a directory consisting of a 'lib' subdirectory with the libcfitsio.a file and a 'include' subdirectory with the fitsio.h file, then you can run the configure script with as single option. For example, if the cfitsio package is installed in this fashion in "/usr/local/ftools' then the configure script option will be...

–with-cfitsio=/usr/local/ftools

If the 'cfitsio' package is not installed in the above manner, then you need to run the configure script with two options, one to specify the include directory and the other to specify the library directory. For example, if the cfitsio package was built in '/home/user/cfitsio' then the two options will be...

–with-cfitsio-include=/home/user/cfitsio –with-cfitsio-libdir=/home/user/cfitsio

You have the option of carrying out the build in a separate directory from the source directory or in hhe same directory as the source. In either case, you need to run the 'configure' script in the directory where the build will occour. So for example, if building in the source directory with the 'cfitsio' directory in '/usr/local/ftools' then the configure coammand should be issued like this...

> ./configure –with-cfitsio=/usr/local/fttools

If you do the build in separate directory from the source, you may need to issue the configure command something like this...

> ../CCfits/configure –with-cfitsio=/usr/local/fttools

The configure script will create the Makefile with the path to the compiler you choose (or GCC by default), and the path to cfitsio package. The configure script has other options, just as the install location. To see this options type

> ./configure –help

2.2.4 Build

Building the C++ shared library will be done automatically by running make without arguments like thus... *On platforms (e.g. OSF) that have a native make program, use of gnu make (gmake) only is supported. On linux, make and gmake are usually synonymous.*

> gmake

This will also build the cookbook executable.

To build on Solaris using the compiler's optimizer, we recommend

> gmake CXXFLAGS=-fast

or

> gmake CXXFLAGS=-O4

[the -fast option will select the best optimization for the particular machine you are compiling on, which may not be the best option if your network is heterogeneous, with SparcStations and Ultra-based workstations of various vintages].

Finally, type in the following to generate the html documentation...

> gmake docs

This will create a directory in your current directory called /html, and all of the html documentation will be created in this directory. The tool 'Doxygen' (http://www.doxygen.org) is required to create the documentation. Executing this won't -quite- give you a manual identical to the one you are reading. However, this option will allow the reader familiar with doxygen to gain additional information about the implementation by playing with the Doxyfile. This may be of utility to users interest in contributing to the library.

2.2.5 Install

To install, type...

> make install

The default install location will be /usr/local/lib for the library and /usr/local/include for the header files. You can change this with the –prefix option when you configure, or with something like...

> make DESTDIR=/usr/local/ftools install

2.2.6 Windows/Visual C++

For compiling the code on a Windows system with MicroSoft Visual C++, take the following steps.

1. Compile the C++ code. Before compiling the C++ code, you may need to edit the file 'fincludes'. This file has the include path to the cfitsio package If the file has the correct path, then open the workspace file 'CCfits.dsw' from within a Visual C++ Studio session and do a build.

2.3 Implementation Notes

This section comments on some of the design decisions for CCfits. We note the role of cfitsio in CCfits as the underlying "engine," the use of the C++ standard library. We also explain some of the choices made for standard library containers in the implementation - all of which is hidden from the user [as it should be]. Finally we explain CCfits' origin as the I/O library for XSPEC12, in the hope that it explains the strengths and weaknesses of the current version and also provides future developers some insights into the design decisions made for one particular data analysis program.

2.3.1 CCfits wraps, not replaces, cfitsio.

Most importantly, the library wraps rather than replaces the use of cfitsio library; it does not perform direct disk I/O. The scheme is designed to retain the well-developed facilities of cfitsio (in particular, the extended file syntax), and make them available to C++ programmers in an OO framework. Some efficiency is lost over a 'pure' C++ FITS library, since the internal C implementation of many functions requires processing if blocks or switch statements that could be recoded in C++ using templates. However, we believe that the current version strikes a resonable compromise between developer time, utility and efficiency.

2.3.2 ANSI C++ and the Standard C++ Library

The implementation of CCfits uses the C++ Standard Library containers and algorithms [also referred to as the Standard Template Library, (STL)] and exception handling. Here is a summary of the rationale behind the implementation decisions made.

- HDUs are contained within a FITS object using a std::multimap<string, HDU*> object.
    1. The map object constructs new array members on first reference
    2. Objects stored in the map are sorted on entry and retrieved efficiently using binary search techniques.
    3. The pointer-to-HDU implementation allows for polymorphism: one set of operations will process all HDU objects within the FITS file
    4. String objects (char*) are represented by the std::string class, which has a rich public interface of search and manipulation facilities.

- Scalar column data [one entry per cell] are implemented using std::vector<T> objects.
- Vector column data [multiple and either fixed or variable numbers of entries per cell] are implemented using std::vector<std::valarray <T> > objects. The std::valarray template is intended for optimized numeric processing. valarrays are have the following desirable features:

  1. they are dynamic, but designed to be allocated in full on construction rather than dynamic resizing during operation: this is, what is usually needed in FITS files.
  2. They have built-in vectorized transcendental functions (e.g. *std::valarray<T> sin(const std::valarray<T>&)*.
  3. They provide std::valarray<T> apply(T f(const T&)) operation, to apply a function f to each element
  4. They provide slicing operations [see the "Getting Started" section for a simple example].

- Exceptions are provided in for by a FitsException hierarchy, which prints out messages on errors and returns control to wherever the exception is caught. Non-zero status values returned by cfitsio are caught by subclass FitsError, which prints the string corresponding to an input status flag. FitsException's other subclasses are thrown on array bounds errors and other programming errors. Rare [we hope] errors that indicate programming flaws in the library throw FitsFatal errors that suggest that the user report the bug.

2.4 Xspec and CCfits

This section is provided for background. Users of CCfits need not read it except to understand how the library was conceived and therefore what its strengths and weaknesses are likely to be in this initial release.

2.4.1 About XSPEC

XSPEC is a general-purpose, multi-mission X-Ray spectral data analysis program which fits data with theoretical models by convolving those models through the instrumental responses. In XSPEC 11.x and all prior versions that use FITS format data, each individual data file format that is supported can have up to 4 anciliary files. That is, for each data file, there can be a response, correction, background and auxiliary response (efficiency) file. Additionally there are table models that read FITS format data. All told, therefore, much duplicated procedural code for reading FITS data can be eliminated by use of the greater encapsulation provided by CCfits. XSPEC's primary need is to read FITS floating point Binary Tables. XSPEC also creates simulated data by convolving users' models with detector responses, so it also has a need for writing tabular data. Images are not used in XSPEC. We have provided a support for image

operations which has undergone limited testing compared to the reading interface for table extensions.

2.4.2 New Data Formats

New formats to be read in XSPEC that are specialized for a particular satellite mission can be supported almost trivially by adding new classes that read data specified with different FITS format files. A single constructor call specifying the required columns and keys is all that is needed to read FITS files, rather than a set of individual cfitsio calls. The library is designed to encourage the "resource acquisition is initialization" model of resource management; CCfits will perform more efficiently if data are read on construction.

2.4.3 Modularity

Third, in an object oriented design, it is possible to make a program only loosely dependent on current implementation assumptions. In XSPEC, data are read as SF and FITS format (of three different varieties) and the user interface is written in tcl/tk. Both of these assumptions could be changed over the future life of the program. Thus the design of XSPEC, and any similar program, consists of defining an abstract DataSet class which has a subclass that uses FITS data. The virtual functions that support reading and writing can easily be overloaded by alternatives to FITS. Thus, the class library specified here fits in with the need for modularity in design.

2.5 Getting Started

The program cookbook.cxx, analogous to the cookbook.c program supplied with cfitsio, was generated to test the correct functioning of the parts of the library and to provide a demonstration of its usage.

The code for cookbook is reproduced here with commentary as worked example of the usage of the library.

2.6 Driver Program

```
// The CCfits headers are expected to be installed in a subdirectory of
// the include path.

// The <CCfits> header file contains all that is neces-
sary to use both the CCfits
// library and the cfitsio library (for example, it includes fit-
sio.h) thus making
// all of cfitsio's macro definitions available.

#ifdef HAVE_CONFIG_H
```

```
#include "config.h"
#endif

// this includes 12 of the CCfits headers and will support all CC-
fits operations.
// the installed location of the library head-
ers is $(ROOT)/include/CCfits

// to use the library either add -I$(ROOT)/include/CCfits or #in-
clude <CCfits/CCfits>
// in the compilation target.

#include <CCfits>

#include <cmath>
    // The library is enclosed in a namespace.

    using namespace CCfits;



int main();
int writeImage();
int writeAscii();
int writeBinary();
int copyHDU();
int selectRows();
int readHeader();
int readImage();
int readTable();

int main()
{


     FITS::setVerboseMode(true);

     try

     {

        if (!writeImage()) std::cerr << " writeImage() \n";
        if (!writeAscii()) std::cerr << " writeAscii() \n";
        if (!writeBinary()) std::cerr << " writeBinary()  \n";
        if (!copyHDU()) std::cerr << " copyHDU() \n";
        if (!readHeader()) std::cerr << " readHeader() \n";
        if (!readImage()) std::cerr << " readImage() \n";
        if (!readTable()) std::cerr << " readTable() \n";
        if (!selectRows()) std::cerr << " selectRows() \n";

     }
     catch (FitsException&)
     // will catch all exceptions thrown by CCfits, including errors
     // found by cfitsio (status != 0)
```

```
    {

       std::cerr << " Fits Exception Thrown by test function \n";

    }
   return 0;
}
```

The simple driver program illustrates the setting of verbose mode for the library, which makes all internal exceptions visible to the programmer. This is primarily for debugging purposes; exceptions are in some cases used to transfer control in common circumstances (e.g. testing whether a file should be created or appended to in write operations). Most of the exceptions will not produce a message unless this flag is set.

Nearly all of the exceptions thrown by CCfits are derived from FitsException, which is caught by reference in the above example. This includes all nonzero status codes returned by cfitsio by the following construct (recall that in the `cfitsio library` nearly all functions return a non-zero status code on error, and have a final argument status of type int):

```
if ( [cfitsio call](args,...,&status)) throw FitsError(status);
```

FitsError, derived from FitsException, uses a cfitsio library call to convert the status code to a string message.

The few exceptions that are not derived from FitsException indicate fatal conditions implying bugs in the library. These print a message suggesting the user contact HEASARC to report the problem.

Note also the lack of statements for closing files in any of the following routines, The destructor (dtor) for the FITS object does this when it falls out of scope. A call

FITS::destroy() throw()

is provided for closing files explicitly; destroy() is also responsible for cleaning up the FITS object and deallocating its resources.

When the data are being read instead of written, the user is expected to copy the data into other program variables [rather than use references to the data contained in the FITS object].

The routines in this program test the following functionality:

writeImage() Writing Primary Images and Image Extensions

writeAscii() Creating and Writing to an Ascii Table Extension

writeBinary() Creating and Writing to a Binary Table Extension

copyHDU() Copying an Extension between Files

selectRows() Selecting Table Data

readHeader() Reading Header information from a HDU

readImage() Reading an Image

readTable() Reading a Table Extension

2.7 Writing Primary Images and Image Extensions

This section of the code demonstrates creation of images. Because every fits file must have a PHDU element, all the FITS constructors (ctors) instantiate a PHDU object. In the case of a new file, the default is to establish an empty HDU with BITPIX = 8 (BYTE_IMG). ***A current limitation of the code is that the data type of the PHDU cannot be replaced after the FITS file is created.*** Arguments to the FITS ctors allow the specification of the data type and the number of axes and their lengths. An image extension of type float is also written by calls in between the writes to the primary header demonstrating switch between HDUs during writes.

Note that in the example below data of type *float* is written to an image of type *unsigned int*, demonstrating both implicit type conversion and the cfitsio extension to unsigned data.

User keywords can be added to the PHDU after successful construction and these will both be accessible as container contents in the in-memory FITS object as well as being written to disk by cfitsio.

Images are represented by the standard library valarray template class which supports vectorized operations on numeric arrays (e.g. taking the square root of an array) and slicing techniques.

The code below also illustrates use of C++ standard library algorithms, and the facilities provided by the std::valarray class.

```
int writeImage()
{

    // Create a FITS primary array containing a 2-D image
    // declare axis arrays.
    long naxis    =   2;
    long naxes[2] = { 300, 200 };

    // declare auto-pointer to FITS at function scope. En-
sures no resources
    // leaked if something fails in dynamic allocation.
    std::auto_ptr<FITS> pFits(0);

    try
    {
        // overwrite existing file if the file already exists.
```

```
        const std::string fileName("!atestfil.fit");

        // Create a new FITS object, specify-
ing the data type and axes for the primary
        // image. Simultaneously create the corresponding file.

        // this image is unsigned short data, demonstrating the cfit-
sio extension
        // to the FITS standard.

        pFits.reset( new FITS(fileName , USHORT_IMG , naxis , naxes ) );
    }
    catch (FITS::CantCreate)
    {
          // ... or not, as the case may be.
          return -1;
    }

    // references for clarity.

    long& vectorLength = naxes[0];
    long& numberOfRows = naxes[1];
    long nelements(1);


    // Find the total size of the array.
    // this is a little fancier than necessary ( It's only
    // calculating naxes[0]*naxes[1]) but it demonstrates  use of the
    // C++ standard library accumulate algorithm.

    nelements = std::accumulate(&naxes[0],&naxes[naxis],1,std::multiplies<long>());

    // create a new image extension with a 300x300 array contain-
ing float data.

    std::vector<long> extAx(2,300);
    string newName ("NEW-EXTENSION");
    ExtHDU* imageExt = pFits->addImage(newName,FLOAT_IMG,extAx);

    // create a dummy row with a ramp. Create an ar-
ray and copy the row to
    // row-sized slices. [also demonstrates the use of valar-
ray slices].
    // also demonstrate implicit type conversion when writ-
ing to the image:
    // input array will be of type float.

    std::valarray<int> row(vectorLength);
    for (long j = 0; j < vectorLength; ++j) row[j] = j;
    std::valarray<int> array(nelements);
    for (int i = 0; i < numberOfRows; ++i)
    {
        array[std::slice(vectorLength*static_cast<int>(i),vectorLength,1)] = row + i;
    }
```

```
    // create some data for the image extension.
    long extElements = std::accumulate(extAx.begin(),extAx.end(),1,std::multiplies<long>());
    std::valarray<float> ranData(extElements);
    const float PIBY (M_PI/150.);
    for ( int jj = 0 ; jj < extElements ; ++jj)
    {
            float arg = PIBY*jj;
            ranData[jj] = std::cos(arg);
    }

    long  fpixel(1);

    // write the image extension data: also demonstrates switch-
ing between
    // HDUs.
    imageExt->write(fpixel,extElements,ranData);

    //add two keys to the primary header, one long, one complex.

    long exposure(1500);
    std::complex<float> omega(std::cos(2*M_PI/3.),std::sin(2*M_PI/3));
    pFits->pHDU().addKey("EXPOSURE", exposure,"Total Exposure Time");
    pFits->pHDU().addKey("OMEGA",omega," Complex cube root of 1 ");


    // The function PHDU& FITS::pHDU() returns a reference to the ob-
ject representing
    // the primary HDU; PHDU::write( <args> ) is then used to write the data.

    pFits->pHDU().write(fpixel,nelements,array);


    // PHDU's friend ostream operator. Doesn't print the entire ar-
ray, just the
    // required & user keywords, and is provided largely for test-
ing purposes [see
    // readImage() for an example of how to output the image ar-
ray to a stream].

    std::cout << pFits->pHDU() << std::endl;

    return 0;
}
```

2.8 Creating and Writing to an Ascii Table Extension

In this section of the program we create a new Table extension of type AsciiTbl, and write three columns with 6 rows. Then we add another copy of the data two rows down (starting from row 3) thus overwriting values and creating new rows. We test the use of null values, and writing a date string. Implicit data conversion, as illustrated for images above, is supported. However, writing numeric data as character data, supported by

cfitsio, is ***not*** supported by CCfits.

Note the basic pattern of CCfits operations: they are performed on an object of type FITS. Access to HDU extension is provided by FITS:: member functions that return references or pointers to objects representing HDUs. Extension are never created directly (all extension ctors are protected), but only through the functions FITS::addTable and FITS::addImage which add extensions to an existing FITS object, performing the necessary cfitsio calls.

The FITS::addTable function takes as one of its last arguments a HDU Type parameter, which needs to be AsciiTbl or BinTbl. The default is to create a BinTable (see next function).

Similarly, access to column data is provided through the functions *ExtHDU::Column*, which return references to columns specified by name or index number - see the documentation for the class ExtHDU for details.

addTable returns a pointer to Table, which is the abstract immediate superclass of the concrete classes AsciiTable and BinTable, whereas addImage returns a pointer to ExtHDU, which is the abstract base class of all FITS extensions. These base classes implement the public interface necessary to avoid the user of the library needing to downcast to a concrete type.

```
int writeAscii ()

    //*************************************************************
    // Create an ASCII Table extension contain-
ing 3 columns and 6 rows *
    //*************************************************************
{
    // declare auto-pointer to FITS at function scope. En-
sures no resources
    // leaked if something fails in dynamic allocation.
    std::auto_ptr<FITS> pFits(0);

    try
    {


        const std::string fileName("atestfil.fit");

        // append the new extension to file created in previous func-
tion call.
        // CCfits writing constructor.

        // if this had been a new file, then the follow-
ing code would create
        // a dummy primary array with BITPIX=8 and NAXIS=0.


        pFits.reset( new FITS(fileName,Write) );
    }
    catch (CCfits::FITS::CantOpen)
```

```
    {
          // ... or not, as the case may be.
          return -1;
    }

    unsigned long rows(6);
    string hduName("PLANETS_ASCII");
    std::vector<string> colName(3,"");
    std::vector<string> colForm(3,"");
    std::vector<string> colUnit(3,"");

    colName[0] = "Planet";
    colName[1] = "Diameter";
    colName[2] = "Density";

    colForm[0] = "a8";
    colForm[1] = "i6";
    colForm[2] = "f4.2";

    colUnit[0] = "";
    colUnit[1] = "km";
    colUnit[2] = "g/cm^-3";

    std::vector<string> planets(rows);

    const char *planet[] = {"Mercury", "Venus", "Earth",
                            "Mars","Jupiter","Saturn"};
    const char *mnemoy[] = {"Many", "Volcanoes", "Erupt",
                            "Mulberry","Jam","Sandwiches","Under",
                                "Normal","Pressure"};

    long diameter[] = {4880,    12112,    12742,   6800,  143000,   121000};
    float density[]  = { 5.1,     5.3,      5.52,   3.94,    1.33,    0.69};

    // append a new ASCII table to the fits file. Note that the user
    // cannot call the Ascii or Bin Table constructors directly as they
    // are protected.

    Table* newTable = pFits->addTable(hduName,rows,colName,colForm,colUnit,AsciiTbl);

    for (size_t j = 0; j < rows; ++j) planets[j] = string(planet[j]);

    // Table::column(const std::string& name) returns a refer-
ence to a Column object

    try
    {

        newTable->column(colName[0]).write(planets,1);
        newTable->column(colName[1]).write(diameter,rows,1);
        newTable->column(colName[2]).write(density,rows,1);

    }
    catch (FitsException&)
    {
```

```
        // ExtHDU::column could in principle throw a NoSuchColumn ex-
ception,
        // or some other fits error may ensue.
        std::cerr << " Error in writing to columns -
check e.g. that columns of specified name "
                      << " exist in the extension \n";

    }

    //  FITSUtil::auto_array_ptr<T> is provided to counter re-
source leaks that
    //  may arise from C-arrays. It is a std::auto_ptr<T> ana-
log that calls
    //  delete[] instead of delete.

    FITSUtil::auto_array_ptr<long>  pDiameter(new long[rows]);
    FITSUtil::auto_array_ptr<float>  pDensity(new float[rows]);
    long* Cdiameter(pDiameter.get());
    float*  Cdensity(pDensity.get());

    Cdiameter[0] = 4880; Cdiameter[1] = 12112; Cdiame-
ter[2] = 12742; Cdiameter[3] = 6800;
    Cdiameter[4] = 143000; Cdiameter[5] = 121000;

    Cdensity[0] = 5.1; Cdensity[1] = 5.3; Cdensity[2] = 5.52; Cden-
sity[3] = 3.94; Cdensity[4] = 1.33;
    Cdensity[5] = 0.69;

    // this << operator outputs everything that has been read.

    std::cout << *newTable << std::endl;

    pFits->pHDU().addKey("NEWVALUE",42," Test of adding keyword to dif-
ferent extension");

    // demonstrate increaing number of rows and null values.
    long ignoreVal(12112);
    long nullNumber(-999);
    try
    {
        // add a TNULLn value to column 2.
        newTable->column(colName[1]).addNullValue(nullNumber);
        // test that writing new data properly expands the num-
ber of rows
        // in both the file]).write(planets,rows-3);
        newTable->column(colName[2]).write(density,rows,rows-3);
        // test the undefined value functionality. Undefineds are re-
placed on
        // disk but not in the memory copy.
        newTable->column(colName[1]).write(diameter,rows,rows-
3,&ignoreVal);
    }
    catch (FitsException&)
    {
            // this time we're going to ignore prob-
```

```
lems in these operations

    }

    // output header information to check that everything we did so far
    // hasn't corrupted the file.

    std::cout << pFits->pHDU() << std::endl;


    std::vector<string> mnemon(9);
    for (size_t j = 0; j < 9; ++j) mnemon[j] = string(mnemoy[j]);

    // Add a new column of string type to the Table.
    // type,  columnName, width, units. [optional - decimals, col-
umn number]
    // decimals is only relevant for floating-
point data in ascii columns.
    newTable->addColumn(Tstring,"Mnemonic",10," words ");
    newTable->column("Mnemonic").write(mnemon,1);

    // write the data string.
    newTable->writeDate();

    // and see if it all worked right.
    std::cout << *newTable << std::endl;

    return 0;
}
```

2.9 Creating and Writing to a Binary Table Extension

The Binary Table interface is more complex because there is an additional parameter, the vector size of each 'cell' in the table, the need to support variable width columns, and the desirability of supporting the input of data in various formats.

The interface supports writing to vector tables the following data structures: C-arrays (T*), std::vector<T> objects, std::valarray<T> objects, and std::vector<valarray<T> >. The last of these is the internal representation of the data.

The function below exercises the following functionality:

- Create a BinTable extension
- Write vector rows to the table
- Insert table rows
- Write complex data to both scalar and vector columns.
- Insert Table columns
- Delete Table rows

- Write HISTORY and COMMENT cards to the Table

```
int writeBinary ()

    //**************************************************************************
    // Create a BINARY table extension and write and manipulate vec-
tor rows
    //**************************************************************************
{
    std::auto_ptr<FITS> pFits(0);

    try
    {

        const std::string fileName("atestfil.fit");
        pFits.reset( new FITS(fileName,Write) );
    }
    catch (CCfits::FITS::CantOpen)
    {
          return -1;
    }


    unsigned long rows(3);
    string hduName("TABLE_BINARY");
    std::vector<string> colName(7,"");
    std::vector<string> colForm(7,"");
    std::vector<string> colUnit(7,"");


    colName[0] = "numbers";
    colName[1] = "sequences";
    colName[2] = "powers";
    colName[3] = "big-integers";
    colName[4] = "dcomplex-roots";
    colName[5] = "fcomplex-roots";
    colName[6] = "scalar-complex";

    colForm[0] = "8A";
    colForm[1] = "20J";
    colForm[2] = "20D";
    colForm[3] = "20V";
    colForm[4] = "20M";
    colForm[5] = "20C";
    colForm[6] = "1M";

    colUnit[0] = "magnets";
    colUnit[1] = "bulbs";
    colUnit[2] = "batteries";
    colUnit[3] = "mulberries";
    colUnit[4] = "";
    colUnit[5] = "";
    colUnit[6] = "pico boo";

    std::vector<string> numbers(rows);
```

```
    const string num("NUMBER-");
    for (size_t j = 0; j < rows; ++j)
    {
#ifdef HAVE_STRSTREAM
        std::ostrstream pStr;
#else
        std::ostringstream pStr;
#endif
        pStr << num << j+1;
        numbers[j] = string(pStr.str());
    }

    const size_t OFFSET(20);

    // write operations take in data as valarray<T>, vector<T> , and
    // vector<valarray<T> >, and T* C-arrays. Create arrays to exer-
cise the C++
    // containers. Check complex I/O for both float and double com-
plex types.

    std::valarray<long> sequence(60);
    std::vector<long> sequenceVector(60);
    std::vector<std::valarray<long> > sequenceVV(3);


    for (size_t j = 0; j < rows; ++j)
    {

        sequence[OFFSET*j] = 1 + j;
        sequence[OFFSET*j+1] = 1 + j;
        sequenceVector[OFFSET*j] = sequence[OFFSET*j];
        sequenceVector[OFFSET*j+1] = sequence[OFFSET*j+1];
        // generate Fibonacci numbers.
        for (size_t i = 2; i < OFFSET; ++i)
        {
                size_t elt (OFFSET*j +i);
                sequence[elt] = sequence[elt-1] + sequence[elt - 2];
                sequenceVector[elt] = sequence[elt] ;
        }
        sequenceVV[j].resize(OFFSET);
        sequenceVV[j] = sequence[std::slice(OFFSET*j,OFFSET,1)];

    }


    std::valarray<unsigned long> unsignedData(60);
    unsigned long base (1 << 31);
    std::valarray<double> powers(60);
    std::vector<double> powerVector(60);
    std::vector<std::valarray<double> > powerVV(3);
    std::valarray<std::complex<double> > croots(60);
    std::valarray<std::complex<float> > fcroots(60);
    std::vector<std::complex<float> > fcroots_vector(60);
    std::vector<std::valarray<std::complex<float> > > fcrootv(3);
```

```
// create complex data as 60th roots of unity.
double PIBY = M_PI/30.;

for (size_t j = 0; j < rows; ++j)
{
    for (size_t i = 0; i < OFFSET; ++i)
    {
            size_t elt (OFFSET*j+i);
            unsignedData[elt] = sequence[elt];
            croots[elt] = std::complex<double>(std::cos(PIBY*elt),std::sin(PIBY*elt));
            fcroots[elt] = std::complex<float>(croots[elt].real(),croots[elt].imag());
            double x = i+1;
            powers[elt] = pow(x,j+1);
            powerVector[elt] = powers[elt];
    }
    powerVV[j].resize(OFFSET);
    powerVV[j] = powers[std::slice(OFFSET*j,OFFSET,1)];
}

FITSUtil::fill(fcroots_vector,fcroots[std::slice(0,20,1)]);

unsignedData += base;
// syntax identical to Binary Table

Table* newTable = pFits->addTable(hduName,rows,colName,colForm,colUnit);

// numbers is a scalar column

newTable->column(colName[0]).write(numbers,1);

// write valarrays to vector column: note signature change
newTable->column(colName[1]).write(sequence,rows,1);
newTable->column(colName[2]).write(powers,rows,1);
newTable->column(colName[3]).write(unsignedData,rows,1);
newTable->column(colName[4]).write(croots,rows,1);
newTable->column(colName[5]).write(fcroots,rows,3);
newTable->column(colName[6]).write(fcroots_vector,1);
// write vectors to column: note signature change

newTable->column(colName[1]).write(sequenceVector,rows,4);
newTable->column(colName[2]).write(powerVector,rows,4);

std::cout << *newTable << std::endl;

for (size_t j = 0; j < 3 ;  ++j)
{
        fcrootv[j].resize(20);
        fcrootv[j] = fcroots[std::slice(20*j,20,1)];
}

// write vector<valarray> object to column.

newTable->column(colName[1]).writeArrays(sequenceVV,7);
newTable->column(colName[2]).writeArrays(powerVV,7);
```

```
    // create a new vector column in the Table


    newTable->addColumn(Tfloat,"powerSeq",20,"none");

    // add data entries to it.

    newTable->column("powerSeq").writeArrays(powerVV,1);
    newTable->column("powerSeq").write(powerVector,rows,4);
    newTable->column("dcomplex-roots").write(croots,rows,4);
    newTable->column("powerSeq").write(sequenceVector,rows,7);


    std::cout << *newTable << std::endl;

    // delete one of the original columns.



    newTable->deleteColumn(colName[2]);


    // add a new set of rows starting after row 3. So we'll have 14 with
    // rows 4,5,6,7,8 blank

    newTable->insertRows(3,5);

    // now, in the new column, write 3 rows (se-
quenceVV.size() = 3). This
    // will place data in rows 3,4,5 of this column,overwriting them.

    newTable->column("powerSeq").writeArrays(sequenceVV,3);
    newTable->column("fcomplex-roots").writeArrays(fcrootv,3);

    // delete 3 rows starting with row 2. A Table:: method, so the same
    // code is called for all Table objects. We should now have 11 rows.

    newTable->deleteRows(2,3);

    //add a history string. This function call is in HDU:: so is identical
    //for all HDUs

    string hist("This file was created for testing CC-
fits write functionality");
    hist += " it serves no other useful purpose. This particu-
lar part of the file was ";
    hist += " constructed to test the writeHistory() and writeCom-
ment() functionality" ;


    newTable->writeHistory(hist);
```

```
    // add a comment string. Use std::string method to change the text in the message
    // and write the previous junk as a comment.

    hist.insert(0, " COMMENT TEST ");

    newTable->writeComment(hist);

    // ... print the result.

    std::cout << *newTable << std::endl;

    return 0;
}
```

2.10 Copying an Extension between Files

Copying extensions from one fits file to another is very straightforward. A complication arises, however, because CCfits requires every FITS object to correspond to a conforming FITS file once constructed. Thus we provide a custom constructor which copies the primary HDU of a "source" FITS file into a new file. Subsequent extensions can be copied by name or extension number as illustrated below.

Note that the simple call

```
FITS::FITS(const std::string& filename)
```

Reads the headers for all of the extensions in the file, so that after the FITS object corresponding to *infile* in the following code is instantiated, all extensions are recognized [read calls are also provided to read only specific HDUs - see below].

In the example code below, the file outFile is written straight to disk. Since the code never requests that the HDUs being written to that file are read, the user needs to add statements to do this after the copy is complete.

```
int copyHDU()
{
    //*****************************************************************
    // copy the 1st and 3rd HDUs from the input file to a new FITS file
    //*****************************************************************


    const string inFileName("atestfil.fit");
    const string outFileName("btestfil.fit");

    int status(0);

    status = 0;

    remove(outFileName.c_str());       // Delete old file if it al-
ready exists
```

```
    // open the existing FITS file
    FITS inFile(inFileName);

    // custom constructor FITS::FITS(const string&, const FITS&) for
    // this particular task.

    FITS outFile(outFileName,inFile);

    // copy extension by number...
    outFile.copy(inFile.extension(2));

    // copy extension by name...
    outFile.copy(inFile.extension("TABLE_BINARY"));

   return 0;

 }
```

2.11 Selecting Table Data

This function demonstrates the operation of filtering a table by selecting rows that satisfy a condition and writing them to a new file, or overwriting a table with the filtered data. A third mode, where a filtered dataset is appended to the file containing the source data, will be available shortly, but is currently not supported by cfitsio.

The expression syntax for the conditions that may be applied to table data are described in the `cfitsio manual.` In the example below, we illustrate filtering with a boolean expression involving one of the columns.

The two flags at the end of the call to FITS::filter are an ‘overwrite’ flag - which only has meaning if the inFile and outFile are the same, and a ’read’ flag. overwrite defaults to true. The second flag is a ’read’ flag which defaults to false. When set true the user has immediate access to the filtered data.

```
int selectRows()
 {
        const string inFile("atestfil.fit");
        const string outFile("btestfil.fit");
        const string newFile("ctestfil.fit");
        remove(newFile.c_str());

        // test 1: write to a new file
        std::auto_ptr<FITS> pInfile(new FITS(inFile,Write,string("PLANETS_ASCII")));
        FITS* infile(pInfile.get());
        std::auto_ptr<FITS> pNewfile(new FITS(newFile,Write));
        ExtHDU& source = infile->extension("PLANETS_ASCII");
        const string expression("DENSITY > 3.0");


        Table& sink1 = pNewfile->filter(expression,source,false,true);
```

```
        std::cout << sink1 << std::endl;

        // test 2: write a new HDU to the current file, over-
write false, read true.
        // AS OF 7/2/01 does not work because of a bug in cfit-
sio, but does not
        // crash, simply writes a new header to the file with-
out also writing the
        // selected data.
        Table& sink2 = infile->filter(expression,source,false,true);

        std::cout << sink2 << std::endl;

        // reset the source file back to the extension in question.
        source = infile->extension("PLANETS_ASCII");

        // test 3: overwrite the current HDU with filtered data.

        Table& sink3 = infile->filter(expression,source,true,true);

        std::cout << sink3 << std::endl;


        return 0;
 }
```

2.12 Reading Header information from a HDU

This function demonstrates selecting one HDU from the file, reading the header information and printing out the keys that have been read and the descriptions of the columns.

The readData flag is by default false (see below for the alternative case), which means that the data in the column is not read.

```
int readHeader()
 {


        const string SPECTRUM("SPECTRUM");

        // read a particular HDU within the file. This call reads just the header
        // information from SPECTRUM

        std::auto_ptr<FITS> pInfile(new FITS("file1.pha",Read,SPECTRUM));

        // define a reference for clarity. (std::auto_ptr<T>::get re-
turns a pointer

        ExtHDU& table = pInfile->extension(SPECTRUM);
```

```
        // read all the keywords, excluding those associ-
ated with columns.

        table.readAllKeys();

        // print the result.


        std::cout << table << std::endl;


       return 0;
}
```

2.13 Reading an Image

Image reading calls are made very simple: the FITS object is created with the readDataFlag set to true, and reading is done on construction. The following call

image.read(contents)

calls

PHDU::read(std::valarray<S>& image).

This copies the entire image from the FITS object into the std::valarray object contents, sizing it as necessary. PHDU::read() and ExtHDU::read() [for image extensions] take a range of arguments that support (a) reading the entire image - as in this example; (b) sections of an image starting from a given pixel; (c) rectangular subsets. See the class references for PHDU and ExtHDU for details.

```
int readImage()
{
       std::auto_ptr<FITS> pInfile(new FITS("atestfil.fit",Read,true));

       PHDU& image = pInfile->pHDU();

       std::valarray<unsigned long>  contents;

       // read all user-specifed, coordinate, and check-
sum keys in the image
       image.readAllKeys();

       image.read(contents);

       // this doesn't print the data, just header info.
       std::cout << image << std::endl;

       long ax1(image.axis(0));
       long ax2(image.axis(1));
```

```
        for (long j = 0; j < ax2; j+=10)
        {
                std::ostream_iterator<short> c(std::cout,"\t");
                std::copy(&contents[j*ax1],&contents[(j+1)*ax1-1],c);
                std::cout << '\n';
        }

        std::cout << std::endl;
        return 0;
}
```

2.14 Reading a Table Extension

Reading table data is similarly straightforward (unsurprisingly, because this application is exactly what CCfits was designed to do easily in the first place).

The two extensions are read on construction, including all the column data [readDataFlag == true] and then printed.

Note that if the data are read as part of the construction, then CCfits uses the row-optimization techniques describe in chapter 13 of the cfitsio manual; a chunk of data equal to the size of the available buffer space is read from contiguous disk blocks and transferred to memory storage, as opposed to each column being read in turn. Thus the most efficient way of reading files is to acquire the data on construction.

```
 int readTable()
 {
        // read a table and explicitly read se-
lected columns. To read instead all the
        // data on construction, set the last argu-
ment of the FITS constructor
        // call to 'true'. This functionality was tested in the last release.
        std::vector<string> hdus(2);
        hdus[0] = "PLANETS_ASCII";
        hdus[1] = "TABLE_BINARY";


        std::auto_ptr<FITS> pInfile(new FITS("atestfil.fit",Read,hdus,false));

        ExtHDU& table = pInfile->extension(hdus[1]);



        std::vector < valarray <int > > pp;
        table.column("powerSeq").readArrays( pp, 1,3 );

        std::vector < valarray <std::complex<double> > > cc;
        table.column("dcomplex-roots").readArrays( cc, 1,3 );

        std::valarray < std::complex<float> > ff;
        table.column("fcomplex-roots").read( ff, 4 );

        std::cout << pInfile->extension(hdus[0]) << std::endl;
```

```
        std::cout << pInfile->extension(hdus[1]) << std::endl;

        return 0;
}
```

2.15 What's Present, What's Missing

Functionality missing from the initial release falls into two categories: facilities that are designed to be supported but are incompletely implemented, and facilities for which there is no support at present. Functions not supported are, of course, accessible through standard calls to cfitsio. However, calls that modify the data that are not supported by the CCfits library will cause differences between the FITS file and the FITS object. Modifying the fits file using the CCfits interface after using cfitsio calls will likely lead to unexpected results.

In addition, it has not been possible to exercise the entire library before release, particularly because of the many functions implemented as member templates. These are created when an application is linked to the library using either g++ and SunPro CC. Much of the read interface for Table extensions, however, has been tested thoroughly during the development of XSPEC.

Most of the functionality of cfitsio described in Chapter 5 of the cfitsio manual is present, although CCfits is designed to provide atomic read/write operations rather than primitive file manipulation. For example, opening and creating FITS files are private operations which are called by reading and writing constructors. Similarly, errors are treated by C++ exception handling rather than returning status codes, and moving between HDUs within a file is a primitive rather than an atomic operation [in CCfits, operations typically call an internal HDU::makeThisCurrent() call on a specific table or image extension, and then perform the requested read/write operation].

Read/Write operations for keys (in the HDU class) are provided; these implement calls to fits_read_key and fits_update_key respectively. In the case of keywords, which have one of five data types (Integer, Logical, String, Floating and Complex) the user is expected to know the data type of the keyword value in this release. In reading image and table data, intrinsic type conversions are performed as in cfitsio with the exception that reading numeric data into character data is **not** supported. There is an extensive set of member functions supporting equivalents of most of cfitsio's read/write operations: the classes PHDU [primary HDU] and ExtHDU [with subclasses template <typename T> ImageExt<T>], provide multiple overloaded versions of read and write functions. The Column class, instances of which can be held in Table instances [with subclasses AsciiTable and BinTable] has also an extensive set of read/write operations.

A special constructor is provided which creates a new file with the Primary HDU of a source file. A FITS::copy(const HDU&) function copies HDUs from one file into another. Support for filtering table rows by expression is provided by a FITS::filter(...) call which may be used to create a new filtered file or overwrite an existing HDU (see

cfitsio manual section 5.6.4).

Functions are provided for adding and deleting columns, and inserting and deleting rows in tables.

HDU objects also have functions to implement writing of history, comment and date keys.

2.15.1 What's Not Present

The coordinate library manipulations [cfitsio manual chapter 7] are not supported.

The iterator work functions [cfitsio manual chapter 6] are not supported. Many of the functions provided are easier to implement using the properties of the standard library, since the standard library containers either allow vectorized operations (in the case of valarrays) or standard library algorithms that take iterators or pointers. In some ways the fits_iterate_data function provide an alternative, approach to the same need for encapsulation addressed by CCfits.

The hierarchical grouping routines are not supported.

The extended file name syntax described in chapters 10 and 11 of the cfitsio manual has not been explicitly tested but should work in that qualified file name strings are passed directly to cfitsio. Template filenames if appearing in parentheses as part of the file name string should likewise be passed directly to cfitsio by CCfits. Failures to open or create the filtered data specified by this syntax will throw either a FITS::CantOpen or FITS::CantCreate exception. Explicit opening of in-memory data sets as described in Chapter 9 of the manual is *not* supported since none of the FITS constructors call the appropriate cfitsio primitives.

2.16 Todo List

Class CCfits::PHDU Implement functions that allow replacement of the primary image

Member CCfits::FITS::addImage(const string &hduName, int bpix, std::vector< long > &naxes, int version=1) Add a function for replacing the primary image

Member CCfits::FITS::addTable(string &hduName, int rows, const std::vector< string > &columnName=std::v the code should one day check that the version keyword is higher than any other versions already added to the FITS object (although cfitsio doesn't do this either).

Member CCfits::AsciiTable::AsciiTable(FITSBase *p, const string &hduName, int rows, const std::vector< strin {enforce equal dimensions for arrays input to AsciiTable, BinTable writing ctor}

Member ITERATORBASE_DEFECT Oded: Write a main program so that HippoDraw can be installed on the desktop. This reqires setting the path to the shared libraries correctly.

Oded: Associate icons with executable and document files. .bmp files for this purpose are in the images subdirectory.

3 CCfits Module Index

3.1 CCfits Modules

Here is a list of all modules:

FITS Exceptions 35

4 CCfits Hierarchical Index

4.1 CCfits Class Hierarchy

This inheritance list is sorted roughly, but not completely, alphabetically:

CCfits::FITSUtil::auto_array_ptr 41

CCfits::FITSUtil::CAarray 47

CCfits::Column 50

CCfits::FITSUtil::CVAarray 63

CCfits::FITSUtil::CVarray 64

CCfits::FITS 75

CCfits::FitsException 88

- **CCfits::Column::InsufficientElements** 98
- **CCfits::Column::InvalidDataType** 99
- **CCfits::Column::InvalidNumberOfRows** 102
- **CCfits::Column::InvalidRowNumber** 103
- **CCfits::Column::InvalidRowParameter** 104

CCfits::Column::NoNullValue 112

CCfits::Column::RangeError 123

CCfits::Column::WrongColumnType 131

CCfits::ExtHDU::WrongExtensionType 132

CCfits::FITS::CantCreate 48

CCfits::FITS::CantOpen 49

CCfits::FITS::NoSuchHDU 115

CCfits::FITS::OperationNotSupported 117

CCfits::FitsError 87

CCfits::HDU::InvalidExtensionType 100

CCfits::HDU::InvalidImageDataType 101

CCfits::HDU::NoNullValue 111

CCfits::HDU::NoSuchKeyword 116

CCfits::Table::NoSuchColumn 113

CCfits::FitsFatal 89

CCfits::HDU 90

CCfits::ExtHDU 65

CCfits::Table 124

CCfits::AsciiTable 37

CCfits::BinTable 44

CCfits::PHDU 118

CCfits::Keyword 105

CCfits::FITSUtil::MatchName 108

CCfits::FITSUtil::MatchNum 109

CCfits::FITSUtil::MatchPtrName 109

CCfits::FITSUtil::MatchPtrNum 110

CCfits::FITSUtil::MatchType 110

CCfits::FITSUtil::MatchType::UnrecognizedType 130

5 CCfits Compound Index

5.1 CCfits Compound List

Here are the classes, structs, unions and interfaces with brief descriptions:

CCfits::AsciiTable (Class Representing Ascii Table Extensions) 37

CCfits::FITSUtil::auto_array_ptr (A class that mimics the std:: library auto_ptr class, but works with arrays) 41

CCfits::BinTable (Class Representing Binary Table Extensions. Contains columns with scalar or vector row entries) 44

CCfits::FITSUtil::CAarray (Function object returning C array from a valarray. see CVarray for details) 47

CCfits::FITS::CantCreate (Thrown on failure to create new file) 48

CCfits::FITS::CantOpen (Thrown on failure to open existing file) 49

CCfits::Column (Abstract base class for Column objects) 50

CCfits::FITSUtil::CVAarray (Function object returning C array from a vector of valarrays. see CVarray for details) 63

CCfits::FITSUtil::CVarray (Function object class for returning C arrays from standard library objects used in the FITS library implementation) 64

CCfits::ExtHDU (Base class for all FITS extension HDUs, i.e. Image Extensions and Tables) 65

CCfits::FITS (Memory object representation of a disk FITS file) 75

CCfits::FitsError (FitsError is the exception thrown by non-zero cfitsio status codes) 87

CCfits::FitsException (FitsException is the base class for all exceptions thrown by this library) 88

CCfits::FitsFatal ([potential] base class for exceptions to be thrown on internal library error) 89

CCfits::HDU (Base class for all HDU [Header-Data Unit] objects) 90

CCfits::Column::InsufficientElements (Exception thrown if the data supplied for a write operation is less than declared) 98

CCfits::Column::InvalidDataType (Exception thrown for invalid data type inputs) 99

CCfits::HDU::InvalidExtensionType (Exception to be thrown if user requests extension type that can not be understood as ImageExt, AsciiTable or BinTable) 100

CCfits::HDU::InvalidImageDataType (Exception to be thrown if user requests creation of an image of type not supported by cfitsio) 101

CCfits::Column::InvalidNumberOfRows (Exception thrown if user enters a non-positive number for the number of rows to write) 102

CCfits::Column::InvalidRowNumber (Exception thrown on attempting to read a row number beyond the end of a table) 103

CCfits::Column::InvalidRowParameter (Exception thrown on incorrect row writing request) 104

CCfits::Keyword (Abstract base class implementing the common behavior of Keyword objects) 105

CCfits::FITSUtil::MatchName (Predicate for classes that have a name attribute; match input string with instance name) 108

CCfits::FITSUtil::MatchNum (Predicate for classes that have an index attribute; match input index with instance value) 109

CCfits::FITSUtil::MatchPtrName (As for MatchName, only with the input class a pointer) 109

CCfits::FITSUtil::MatchPtrNum (As for MatchNum, only with the input class a pointer) 110

CCfits::FITSUtil::MatchType (Function object that returns the FITS ValueType corresponding to an input intrinsic type) 110

CCfits::HDU::NoNullValue (Exception to be thrown on seek errors for keywords) 111

CCfits::Column::NoNullValue (Exception thrown if a null value is specified without support from existing column header) 112

CCfits::Table::NoSuchColumn (Exception to be thrown on a failure to retrieve a column specified either by name or index number) 113

CCfits::FITS::NoSuchHDU (Exception thrown by HDU retrieval methods) 115

CCfits::HDU::NoSuchKeyword (Exception to be thrown on seek errors for keywords) 116

CCfits::FITS::OperationNotSupported (Thrown for unsupported operations, such as attempted to select rows from an image extension) 117

CCfits::PHDU (Class representing the primary HDU for a FITS file) 118

CCfits::Column::RangeError (Exception to be thrown for inputs that cause range errors in column read operations) 123

CCfits::Table 124

CCfits::FITSUtil::MatchType::UnrecognizedType (Exception thrown by MatchType if it encounters data type incompatible with cfitsio) 130

CCfits::Column::WrongColumnType (Exception thrown on attempting to access a scalar column as vector data) 131

CCfits::ExtHDU::WrongExtensionType (Exception to be thrown on unmatched extension types) 132

6 CCfits Module Documentation

6.1 FITS Exceptions

Compounds

- class CCfits::FITS::CantCreate

  thrown on failure to create new file.

- class CCfits::FITS::CantOpen

  thrown on failure to open existing file.

- class CCfits::FitsError

  FitsError is the exception thrown by non-zero cfitsio status codes.

- class CCfits::FitsException

  FitsException is the base class for all exceptions thrown by this library.

- class CCfits::FitsFatal

  [potential] base class for exceptions to be thrown on internal library error.

- class CCfits::HDU::InvalidExtensionType

  exception to be thrown if user requests extension type that can not be understood as ImageExt, AsciiTable or BinTable.

- class CCfits::HDU::InvalidImageDataType

  exception to be thrown if user requests creation of an image of type not supported by cfitsio.

- class CCfits::HDU::NoNullValue

  exception to be thrown on seek errors for keywords.

- class CCfits::Table::NoSuchColumn

  Exception to be thrown on a failure to retrieve a column specified either by name or index number.

- class CCfits::FITS::NoSuchHDU

  exception thrown by HDU retrieval methods.

- class CCfits::HDU::NoSuchKeyword

  exception to be thrown on seek errors for keywords.

- class CCfits::FITS::OperationNotSupported

  thrown for unsupported operations, such as attempted to select rows from an image extension.

- class CCfits::Column::RangeError

  exception to be thrown for inputs that cause range errors in column read operations.

- class CCfits::FITSUtil::MatchType::UnrecognizedType

  exception thrown by MatchType if it encounters data type incompatible with cfitsio.

- class CCfits::ExtHDU::WrongExtensionType

  Exception to be thrown on unmatched extension types.

7 CCfits Namespace Documentation

7.1 FITSUtil Namespace Reference

FITSUtil is a namespace containing functions used internally by CCfits, but which might be of use for other applications.

7.1.1 Detailed Description

FITSUtil is a namespace containing functions used internally by CCfits, but which might be of use for other applications.

8 CCfits Class Documentation

8.1 CCfits::AsciiTable Class Reference

Class Representing Ascii Table Extensions.

`#include <AsciiTable.h>`

Inheritance diagram for CCfits::AsciiTable::

Public Methods

- virtual AsciiTable* clone (FITSBase *p) const
- virtual void readData (bool readFlag=false, const std::vector< string > &keys=std::vector< string >())
- virtual void addColumn (ValueType type, const string &columnName, long repeatWidth, const string &colUnit=string(""), long decimals=-1, size_t columnNumber=0)

Protected Methods

- AsciiTable (FITSBase *p, const string &hduName=string(""), bool readFlag=false, const std::vector< string > &keys=std::vector< string >(), int version=1)
- AsciiTable (FITSBase *p, const string &hduName, int rows, const std::vector< string > &columnName=std::vector< string >(), const std::vector< string > &columnFmt=std::vector< string >(), const std::vector< string > &columnUnit=std::vector< string >(), int version=1)

- AsciiTable (FITSBase *p, int number)
- ~AsciiTable ()

8.1.1 Detailed Description

Class Representing Ascii Table Extensions.

May only contain columns with scalar row entries and a small range of data types. AsciiTable (re)implements functions prescribed in the Table abstract class. The implementations allow the calling of cfitsio specialized routines for AsciiTable header construction.

Direct instantiation of AsciiTable objects is disallowed: they are created by explicit calls to FITS::addTable(...), FITS::read(...) or internally by one of the FITS ctors on initialization. The default for FITS::addTable is to produce BinTable extensions.

8.1.2 Constructor & Destructor Documentation

8.1.2.1 CCfits::AsciiTable::AsciiTable (FITSBase * *p*, const string & *hduName* = string(""), bool *readFlag* = false, const std::vector< string > & *keys* = std::vector<string>(), int *version* = 1) [protected]

reading constructor: Construct a AsciiTable extension from an extension of an existing disk file.

The Table is specified by name and optional version number within the file. An array of strings representing columns or keys indicates which data are to be read. The column data are only read if readFlag is true. Reading on construction is optimized, so it is more efficient to read data at the point of instantiation. This favours a "resource acquisition is initialization" model of data management.

Parameters:

hduName name of AsciiTable object to be read.

readFlag flag to determine whether to read data on construction

keys (optional) a list of keywords/columns to be read. The implementation will determine which are keywords. If none are specified, the constructor will simply read the header

version (optional) version number. If not specified, will read the first extension that matches hduName.

8.1.2.2 CCfits::AsciiTable::AsciiTable (FITSBase * *p*, const string & *hduName*, int *rows*, const std::vector< string > & *columnName* = std::vector<string>(), const std::vector< string > & *columnFmt* = std::vector<string>(), const std::vector< string > & *columnUnit* = std::vector<string>(), int *version* = 1) [protected]

writing constructor: create new Ascii Table object with the specified columns.

The constructor creates a valid HDU which is ready for Column::write or insertRows operations. The disk FITS file is update accordingly. The data type of each column is determined by the columnFmt argument (TFORM keywords). See cfitsio documentation for acceptable values.

Parameters:

hduName name of AsciiTable object to be written

rows number of rows in the table (NAXIS2)

columnName array of column names for columns to be constructed.

columnFmt array of column formats for columns to be constructed.

columnUnit (optional) array of units for data in columns.

version (optional) version number for HDU.

The dimensions of columnType, columnName and columnFmt must match, although this is not enforced at present.

Todo:

{enforce equal dimensions for arrays input to AsciiTable, BinTable writing ctor}

8.1.2.3 CCfits::AsciiTable::AsciiTable (FITSBase * *p*, int *number*) [protected]

read AsciiTable with HDU number `number` from existing file.

This is used internally by methods that need to access HDUs for which no EXTNAME [or equivalent] keyword exists.

8.1.2.4 CCfits::AsciiTable::~AsciiTable () [protected]

destructor.

8.1.3 Member Function Documentation

8.1.3.1 void CCfits::AsciiTable::addColumn (ValueType *type*, const string & *columnName*, long *repeatWidth*, const string & *colUnit* = string(""), long *decimals* = -1, size_t *columnNumber* = 0) [virtual]

add a new column to an existing table HDU.

Parameters:

type The data type of the column to be added

columnName The name of the column to be added

repeatWidth for a string valued, this is the width of a string. For a numeric column it supplies the vector length of the rows. It is ignored for ascii table numeric data.

colUnit an optional field specifying the units of the data (TUNITn)

decimal optional parameter specifying the number of decimals for an ascii numeric column

columnNumber optional parameter specifying column number to be created. If not specified the column is added to the end. If specified, the column is inserted and the columns already read are reindexed. This parameter is provided as a convenience to support existing code rather than recommended.

Reimplemented from CCfits::ExtHDU.

8.1.3.2 AsciiTable * CCfits::AsciiTable::clone (FITSBase * *p*) const [virtual]

virtual copy constructor.

Reimplemented from CCfits::ExtHDU.

8.1.3.3 void CCfits::AsciiTable::readData (bool *readFlag* = false, const std::vector< string > & *keys* = std::vector<string>()) [virtual]

read columns and keys specified in the input array.

See Table class documentation for further details.

Reimplemented from CCfits::ExtHDU.

The documentation for this class was generated from the following files:

- AsciiTable.h
- AsciiTable.cxx

8.2 CCfits::FITSUtil::auto_array_ptr Class Template Reference

A class that mimics the std:: library auto_ptr class, but works with arrays.

`#include <FITSUtil.h>`

Public Methods

- auto_array_ptr (X *p=0) throw ()
- auto_array_ptr (auto_array_ptr< X > &right) throw ()

- ~auto_array_ptr ()
- void operator= (auto_array_ptr< X > &right)
- X& operator * () throw ()
- X& operator[ ] (size_t i) throw ()
- X operator[ ] (size_t i) const throw ()
- X* get () const
- X* release () throw ()
- X* reset (X *p) throw ()

Static Public Methods

- void remove (X *&x)

8.2.1 Detailed Description

template<typename X> class CCfits::FITSUtil::auto_array_ptr

A class that mimics the std:: library auto_ptr class, but works with arrays.

This code was written by Jack Reeves and first appeared C++ Report, March 1996 edition. Although some authors think one shouldn't need such a contrivance, there seems to be a need for it when wrapping C code.

Usage: replace

float* *f* = new float[200];

with

FITSUtil::auto_array_ptr<float> *f(new float[200])*;

Then the memory will be managed correctly in the presence of exceptions, and delete will be called automatically for *f* when leaving scope.

8.2.2 Constructor & Destructor Documentation

8.2.2.1 template<typename X> CCfits::FITSUtil::auto_array_ptr< X >::auto_array_ptr<X> (X * *p* = 0) throw () [explicit]

constructor. allows creation of pointer to null, can be modified by reset().

8.2.2.2 template<typename X> CCfits::FITSUtil::auto_array_ptr< X >::auto_array_ptr<X> (auto_array_ptr< X > & *right*) throw () [explicit]

copy constructor.

8.2.2.3 template<typename X> CCfits::FITSUtil::auto_array_ptr< X >::~auto_array_ptr<X> ()

destructor.

8.2.3 Member Function Documentation

8.2.3.1 template<typename X> X * CCfits::FITSUtil::auto_array_ptr< X >::get () const

return a token for the underlying content of *this.

8.2.3.2 template<typename X> X & CCfits::FITSUtil::auto_array_ptr< X >::operator * () throw ()

deference operator.

8.2.3.3 template<typename X> void CCfits::FITSUtil::auto_array_ptr< X >::operator= (auto_array_ptr< X > & *right*)

assignment operator: transfer of ownership semantics.

8.2.3.4] template<typename X> X CCfits::FITSUtil::auto_array_ptr< X >::operator[ ] (size_t *i*) const throw ()

return a copy of the ith element of the array.

8.2.3.5] template<typename X> X & CCfits::FITSUtil::auto_array_ptr< X >::operator[ ] (size_t *i*) throw ()

return a reference to the ith element of the array.

8.2.3.6 template<typename X> X * CCfits::FITSUtil::auto_array_ptr< X >::release () throw ()

return underlying content of *this, transferring memory ownership.

8.2.3.7 template<typename X> void CCfits::FITSUtil::auto_array_ptr< X >::remove (X *& *x*) [static]

utility function to delete the memory owned by x and set it to null.

8.2.3.8 template<typename X> X * CCfits::FITSUtil::auto_array_ptr< X >::reset (X * *p*) throw ()

change the content of the auto_array_ptr to p.

The documentation for this class was generated from the following file:

- FITSUtil.h

8.3 CCfits::BinTable Class Reference

Class Representing Binary Table Extensions. Contains columns with scalar or vector row entries.

`#include <BinTable.h>`

Inheritance diagram for CCfits::BinTable::

Public Methods

- virtual BinTable* clone (FITSBase *p) const
- virtual void readData (bool readFlag=false, const std::vector< string > &keys=std::vector< string >())
- virtual void addColumn (ValueType type, const string &columnName, long repeatWidth, const string &colUnit=string(""), long decimals=-1, size_t columnNumber=0)

Protected Methods

- BinTable (FITSBase *p, const string &hduName=string(""), bool readFlag=false, const std::vector< string > &keys=std::vector< string >(), int version=1)
- BinTable (FITSBase *p, const string &hduName, int rows, const std::vector< string > &columnName=std::vector< string >(), const std::vector< string > &columnFmt=std::vector< string >(), const std::vector< string > &columnUnit=std::vector< string >(), int version=1)
- BinTable (FITSBase *p, int number)
- ~BinTable ()

8.3.1 Detailed Description

Class Representing Binary Table Extensions. Contains columns with scalar or vector row entries.

BinTable (re)implements functions prescribed in the Table abstract class. The implementations allow the calling of cfitsio specialized routines for BinTable header construction. functions particular to the BinTable class include those dealing with variable width columns

Direct instantiation of BinTable objects is disallowed: they are created by explicit calls to FITS::addTable(...), FITS::read(...) or internally by one of the FITS ctors on initialization. For addTable, creation of BinTables is the default.

8.3.2 Constructor & Destructor Documentation

8.3.2.1 CCfits::BinTable::BinTable (FITSBase * *p*, const string & *hduName* = string(""), bool *readFlag* = false, const std::vector< string > & *keys* = std::vector<string>(), int *version* = 1) [protected]

reading constructor.

Construct a BinTable extension from an extension of an existing disk file. The Table is specified by name and optional version number within the file. An array of strings representing columns or keys indicates which data are to be read. The column data are only read if readFlag is true. Reading on construction is optimized, so it is more efficient to read data at the point of instantiation. This favours a "resource acquisition is initialization" model of data management.

Parameters:

hduName name of BinTable object to be read.

readFlag flag to determine whether to read data on construction

keys (optional) a list of keywords/columns to be read. The implementation will determine which are keywords. If none are specified, the constructor will simply read the header

version (optional) version number. If not specified, will read the first extension that matches hduName.

8.3.2.2 CCfits::BinTable::BinTable (FITSBase * *p*, const string & *hduName*, int *rows*, const std::vector< string > & *columnName* = std::vector<string>(), const std::vector< string > & *columnFmt* = std::vector<string>(), const std::vector< string > & *columnUnit* = std::vector<string>(), int *version* = 1) [protected]

writing constructor.

The constructor creates a valid HDU which is ready for Column::write or insertRows operations. The disk FITS file is update accordingly. The data type of each column is determined by the columnFmt argument (TFORM keywords). See cfitsio documentation for acceptable values.

Parameters:

hduName name of BinTable object to be written

rows number of rows in the table (NAXIS2)

columnName array of column names for columns to be constructed.

columnFmt array of column formats for columns to be constructed.

columnUnit (optional) array of units for data in columns.

version (optional) version number for HDU.

The dimensions of columnType, columnName and columnFmt must match, but this is not enforced.

8.3.2.3 CCfits::BinTable::BinTable (FITSBase * *p*, int *number*) [protected]

read BinTable with HDU number `number` from existing file represented by fitsfile pointer p.

8.3.2.4 CCfits::BinTable::~BinTable () [protected]

destructor.

8.3.3 Member Function Documentation

8.3.3.1 void CCfits::BinTable::addColumn (ValueType *type*, const string & *columnName*, long *repeatWidth*, const string & *colUnit* = string(""), long *decimals* = -1, size_t *columnNumber* = 0) [virtual]

add a new column to an existing table HDU.

Parameters:

type The data type of the column to be added

columnName The name of the column to be added

repeatWidth for a string valued, this is the width of a string. For a numeric column it supplies the vector length of the rows. It is ignored for ascii table numeric data.

colUnit an optional field specifying the units of the data (TUNITn)

decimal optional parameter specifying the number of decimals for an ascii numeric column

columnNumber optional parameter specifying column number to be created. If not specified the column is added to the end. If specified, the column is inserted and the columns already read are reindexed. This parameter is provided as a convenience to support existing code rather than recommended.

Reimplemented from CCfits::ExtHDU.

8.3.3.2 BinTable * CCfits::BinTable::clone (FITSBase * *p*) const [virtual]

virtual copy constructor.

Reimplemented from CCfits::ExtHDU.

8.3.3.3 void CCfits::BinTable::readData (bool *readFlag* = false, const std::vector< string > & *keys* = std::vector<string>()) [virtual]

read columns and keys specified in the input array.

See Table class documentation for further details.

Reimplemented from CCfits::ExtHDU.

The documentation for this class was generated from the following files:

- BinTable.h
- BinTable.cxx

8.4 CCfits::FITSUtil::CAarray Class Template Reference

function object returning C array from a valarray. see CVarray for details.

`#include <FITSUtil.h>`

Public Methods

- T* operator() (const std::valarray< T > &inArray)

8.4.1 Detailed Description

template<typename T> class CCfits::FITSUtil::CAarray

function object returning C array from a valarray. see CVarray for details.

8.4.2 Member Function Documentation

8.4.2.1 template<typename T> T * CCfits::FITSUtil::CAarray< T >::operator() (const std::valarray< T > & *inArray*)

operator returning C array for use with image data.

The documentation for this class was generated from the following file:

- FITSUtil.h

8.5 CCfits::FITS::CantCreate Class Reference

thrown on failure to create new file.

`#include <FITS.h>`

Inheritance diagram for CCfits::FITS::CantCreate::

Public Methods

- CantCreate (const string &diag, bool silent=false)

8.5.1 Detailed Description

thrown on failure to create new file.

8.5.2 Constructor & Destructor Documentation

8.5.2.1 CCfits::FITS::CantCreate::CantCreate (const string & *diag*, bool *silent* = false)

Exception ctor prefixes the string: "FITS Error: Cannot create file " before specific message.

This exception will be thrown if the user attempts to write to a protected directory or attempts to create a new file with the same name as an existing file without specifying overwrite [overwrite is specified by adding the character '!' before the filename, following the cfitsio convention].

Parameters:

msg A specific diagnostic message, the name of the file that was to be created.

silent if true, print message whether FITS::verboseMode is set or not.

The documentation for this class was generated from the following files:

- FITS.h
- FITS.cxx

8.6 CCfits::FITS::CantOpen Class Reference

thrown on failure to open existing file.

`#include <FITS.h>`

Inheritance diagram for CCfits::FITS::CantOpen::

Public Methods

- CantOpen (const string &diag, bool silent=false)

8.6.1 Detailed Description

thrown on failure to open existing file.

8.6.2 Constructor & Destructor Documentation

8.6.2.1 CCfits::FITS::CantOpen::CantOpen (const string & *diag*, bool *silent* = false)

Exception ctor prefixes the string: "FITS Error: Cannot create file " before specific message.

This exception will be thrown if users attempt to open an existing file for write access to which they do not have permission, or of course if the file does not exist.

Parameters:

diag A specific diagnostic message, the name of the file that was to be created.

silent if true, print message whether FITS::verboseMode is set or not.

The documentation for this class was generated from the following files:

- FITS.h
- FITS.cxx

8.7 CCfits::Column Class Reference

Abstract base class for Column objects.

`#include <Column.h>`

Public Methods

- Column (const Column &right)

- virtual ~Column ()
- virtual void readData (long firstRow, long nelements, long firstElem=1)=0
- int rows () const
- void setDisplay ()
- virtual void setDimen ()
- int index () const
- bool isRead () const
- long width () const
- size_t repeat () const
- bool varLength () const
- double scale () const
- double zero () const
- const string& display () const
- const string& dimen () const
- ValueType type () const
- const std::string& name () const
- const std::string& unit () const
- const std::string& format () const
- template<typename S> void write (const std::vector< S > &indata, long firstRow)
- template<typename S> void write (const std::valarray< S > &indata, long firstRow)
- template<typename S> void write (S *indata, long nRows, long firstRow)
- template<typename S> void write (const std::vector< S > &indata, long firstRow, S *nullValue)
- template<typename S> void write (const std::valarray< S > &indata, long firstRow, S *nullValue)
- template<typename S> void write (S *indata, long nRows, long firstRow, S *nullValue)
- template<typename S> void write (const std::valarray< S > &indata, long nRows, long firstRow)
- template<typename S> void write (const std::vector< S > &indata, long nRows, long firstRow)
- template<typename S> void write (S *indata, long nElements, long nRows, long firstRow)
- template<typename S> void write (const std::valarray< S > &indata, long nRows, long firstRow, S *nullValue)
- template<typename S> void write (const std::vector< S > &indata, long nRows, long firstRow, S *nullValue)
- template<typename S> void write (S *indata, long nElements, long nRows, long firstRow, S *nullValue)
- template<typename S> void write (const std::valarray< S > &indata, const std::vector< long > &vectorLengths, long firstRow)

- template<typename S> void write (const std::vector< S > &indata, const std::vector< long > &vectorLengths, long firstRow)
- template<typename S> void write (S *indata, long nElements, const std::vector< long > &vectorLengths, long firstRow)
- template<typename S> void writeArrays (const std::vector< std::valarray< S > > &indata, long firstRow)
- template<typename S> void writeArrays (const std::vector< std::valarray< S > > &indata, long firstRow, S *nullValue)
- template<typename S> void read (std::vector< S > &vals, long first, long last)
- template<typename S> void read (std::valarray< S > &vals, long first, long last)
- template<typename S> void read (std::valarray< S > &vals, long rows)
- template<typename S> void readArrays (std::vector< std::valarray< S > > &vals, long first, long last)
- template<typename S> void read (std::vector< S > &vals, long first, long last, S *nullValue)
- template<typename S> void read (std::valarray< S > &vals, long first, long last, S *nullValue)
- template<typename S> void read (std::valarray< S > &vals, long rows, S *nullValue)
- template<typename S> void readArrays (std::vector< std::valarray< S > > &vals, long first, long last, S *nullValue)
- template<typename T> void addNullValue (T nullVal)

Protected Methods

- Column (int columnIndex, const string &columnName, ValueType type, const string &format, const string &unit, Table *p, int rpt=1, long w=1, const string &comment="")
- Column (Table *p=0)
- fitsfile* fitsPointer ()
- void makeHDUCurrent ()
- virtual std::ostream& put (std::ostream &s) const
- const std::string& comment () const

8.7.1 Detailed Description

Abstract base class for Column objects.

Columns are the data containers used in FITS tables. Columns of scalar type (one entry per cell) are implemented by the template subclass ColumnData<T>. Columns of vector type (vector and variable rows) are implemented with the template subclass ColumnVectorData<T>. AsciiTables may only contain Columns of type ColumnData<T>, where T is an implemented FITS data type (see the CCfits.h header for a

complete list. This requirement is enforced by ensuring that AsciiTable's addColumn method may only create an AsciiTable compatible column. The ColumnData<T> class stores its data in a std::vector<T> object.

BinTables may contain either ColumnData<T> or ColumnVectorData<T>. For ColumnVectorData, T must be a numeric type: string vectors are handled by ColumnData<T>; string arrays are not supported. The internal representation of the data is a std::vector<std::valarray<T> > object. The std::valarray class is designed for efficient numeric processing and has many vectorized numeric and transcendental functions defined on it.

Member template functions for read/write operations are provided in multiple overloads as the interface to data operations. Implicit data type conversions are supported but where they are required make the operations less efficient. Reading numeric column data as character arrays, supported by cfitsio, is not supported by CCfits.

As a base class, Column provides protected accessor/mutator inline functions to allow only its subclasses to access data members.

8.7.2 Constructor & Destructor Documentation

8.7.2.1 CCfits::Column::Column (const Column & *right*)

copy constructor, used in copying Columns to standard library containers.

The copy constructor is for internal use only: it does not affect the disk fits file associated with the object.

8.7.2.2 CCfits::Column::~Column () [virtual]

destructor.

8.7.2.3 CCfits::Column::Column (int *columnIndex*, const string & *columnName*, ValueType *type*, const string & *format*, const string & *unit*, Table * *p*, int *rpt* = 1, long *w* = 1, const string & *comment* = "") [protected]

new column creation constructor.

This constructor allows the specification of:

Parameters:

columnIndex The column number

columnName The column name, keyword TTYPEn

type used for determining class of T in ColumnData<T>, ColumnVectorData<T>

format the column data format, TFORMn keyword

unit the column data unit, TUNITn keyword

p the Table pointer

rpt (optional) repeat count for the row (== 1 for AsciiTables)

w the row width

comment comment to be added to the header.

8.7.2.4 CCfits::Column::Column (Table * *p* = 0) [protected]

Simple constructor to be called by subclass reading ctors.

8.7.3 Member Function Documentation

8.7.3.1 template<typename T> void CCfits::Column::addNullValue (T *nullVal*)

Set the TNULLn keyword for the column.

Only relevant for integer valued columns, TNULLn is the value used by cfitsio in undefined processing. All entries in the table equal to an input "null value" are set equal to the value of TNULLn. (For floating point columns a system NaN values is used).

8.7.3.2 const std::string & CCfits::Column::comment () const [inline, protected]

retrieve comment for Column.

8.7.3.3 const string & CCfits::Column::dimen () const [inline]

return TDIMn keyword.

represents dimensions of data arrays in vector columns. for scalar columns, returns a default value.

8.7.3.4 const string & CCfits::Column::display () const [inline]

return TDISPn keyword.

TDISPn is suggested format for output of column data.

8.7.3.5 fitsfile * CCfits::Column::fitsPointer () [protected]

fits pointer corresponding to fits file containing column data.

8.7.3.6 const std::string & CCfits::Column::format () const [inline]

return TFORMn keyword.

TFORMn specifies data format stored in disk file.

8.7.3.7 int CCfits::Column::index () const [inline]

get the Column index (the n in TTYPEn etc).

8.7.3.8 bool CCfits::Column::isRead () const [inline]

flag set to true if the entire column data has been read from disk.

8.7.3.9 void CCfits::Column::makeHDUCurrent () [protected]

make HDU containing this the current HDU of the fits file.

8.7.3.10 const std::string & CCfits::Column::name () const [inline]

return name of Column (TTYPEn keyword).

8.7.3.11 std::ostream & CCfits::Column::put (std::ostream & *s*) const [protected, virtual]

internal implementation of << operator.

8.7.3.12 template<typename S> void CCfits::Column::read (std::valarray< S > & *vals*, long *row*, S * *nullValue*)

return a single row of a vector column into a std::valarray, setting undefined values.

8.7.3.13 template<typename S> void CCfits::Column::read (std::valarray< S > & *vals*, long *first*, long *last*, S * *nullValue*)

Retrieve data from a scalar column into a std::valarray, setting undefined values.

Parameters:

vals The output container. The function will resize this as necessary

first,last the span of row numbers to read.

8.7.3.14 template<typename S> void CCfits::Column::read (std::vector< S > & *vals*, long *first*, long *last*, S * *nullValue*)

Retrieve data from a scalar column into a std::vector, setting nullvalue.

As above, only a pointer to a null value is also recognized. If the column is of integer type, then any column value that equals this null value is set equal to the value of the TNULLn keyword. An exception is thrown if TNULLn is not specified. See cfitsio documentation for further details

Parameters:

vals The output container. The function will resize this as necessary

first,last the span of row numbers to read.

nullValue, pointer to integer value regarded as undefined

8.7.3.15 template<typename S> void CCfits::Column::read (std::valarray< S > & *vals*, long *row*)

return a single row of a vector column into a std::valarray.

Parameters:

vals The output valarray object

rows The row number to be retrieved (starting at 1).

8.7.3.16 template<typename S> void CCfits::Column::read (std::valarray< S > & *vals*, long *first*, long *last*)

Retrieve data from a scalar column into a std::valarray.

Parameters:

vals The output container. The function will resize this as necessary

first,last the span of row numbers to read.

8.7.3.17 template<typename S> void CCfits::Column::read (std::vector< S > & *vals*, long *first*, long *last*)

Retrieve data from a scalar column into a std::vector.

This and the following functions perform implicit data conversions. An exception will be thrown if no conversion exists.

Parameters:

vals The output container. The function will resize this as necessary

first,last the span of row numbers to read.

8.7.3.18 template<typename S> void CCfits::Column::readArrays (std::vector< std::valarray< S > > & *vals*, long *first*, long *last*, S * *nullValue*)

return a set of rows of a vector column into a container, setting undefined values.

Parameters:

vals The output container. The function will resize this as necessary

first,last the span of row numbers to read.

8.7.3.19 template<typename S> void CCfits::Column::readArrays (std::vector< std::valarray< S > > & *vals*, long *first*, long *last*)

return a set of rows of a vector column into a vector of valarrays.

Parameters:

vals The output container. The function will resize this as necessary

first,last the span of row numbers to read.

8.7.3.20 void CCfits::Column::readData (long *firstrow* = 1, long *nelements* = 1, long *firstelem* = 1) [pure virtual]

read method.

Parameters:

firstRow The first row to be read

firstElem the number of the element on the first row to start at (ignored for scalar columns)

nRows The number of rows to read

8.7.3.21 size_t CCfits::Column::repeat () const [inline]

get the repeat count for the rows.

8.7.3.22 int CCfits::Column::rows () const

return number of rows in the Column.

8.7.3.23 double CCfits::Column::scale () const [inline]

get TSCALn value.

TSCALn is used to convert a data array represented on disk in integer format as floating. Useful for compact storage of digitized data.

8.7.3.24 void CCfits::Column::setDimen () [inline, virtual]

set the TDIMn keyword.

8.7.3.25 void CCfits::Column::setDisplay ()

set the TDISPn keyword.

8.7.3.26 ValueType CCfits::Column::type () const [inline]

returns the data type of the column.

8.7.3.27 const std::string & CCfits::Column::unit () const [inline]

get units of data in Column (TUNITn keyword).

8.7.3.28 bool CCfits::Column::varLength () const [inline]

boolean, set to true if Column has variable length vector rows.

8.7.3.29 long CCfits::Column::width () const [inline]

return column data width.

8.7.3.30 template<typename S> void CCfits::Column::write (S * *indata*, long *nelements*, const std::vector< long > & *vectorLengths*, long *firstRow*)

write a C-array of values of size nElements into a column with specified number of entries written per row.

8.7.3.31 template<typename S> void CCfits::Column::write (const std::vector< S > & *indata*, const std::vector< long > & *vectorLengths*, long *firstRow*)

write a vector of values into a column with specified number of entries written per row.

8.7.3.32 template<typename S> void CCfits::Column::write (const std::valarray< S > & *indata*, const std::vector< long > & *vectorLengths*, long *firstRow*)

write a valarray of values into a column with specified number of entries written per row.

Data are written into vectorLengths.size() rows, with vectorLength[n] elements written to row n+firstRow -1. Although clearly designed for wrapping calls to multiple

variable-width vector column rows, the code is written to write a variable number of elements to fixed-width column rows (cfitsio requires these operations to be done on a row-by-row basis).

Since cfitsio does not support null value processing for variable width columns this function and its variants do not have version which process undefined values

Parameters:

indata The data to be written

vectorLengths the number of elements to write to each successive row.

firstRow the first row to be written.

8.7.3.33 template<typename S> void CCfits::Column::write (S * *indata*, long *nelements*, long *nRows*, long *firstRow*, S * *nullValue*)

write a C array of values into a range of rows of a vector column, processing undefined values.

8.7.3.34 template<typename S> void CCfits::Column::write (const std::vector< S > & *indata*, long *nRows*, long *firstRow*, S * *nullValue*)

write a vector of values into a range of rows of a vector column, processing undefined values.

see valarray version for details.

8.7.3.35 template<typename S> void CCfits::Column::write (const std::valarray< S > & *indata*, long *nRows*, long *firstRow*, S * *nullValue*)

write a valarray of values into a range of rows of a vector column.

see version without undefined processing for details.

8.7.3.36 template<typename S> void CCfits::Column::write (S * *indata*, long *nelements*, long *nRows*, long *firstRow*)

write a C array of values into a range of rows of a vector column.

Details are as for vector input; only difference is the need to supply the size of the C-array.

Parameters:

indata The data to be written.

nElements The size of indata

nRows the number of rows to which to write the data.

firstRow The first row to be written

8.7.3.37 template<typename S> void CCfits::Column::write (const std::vector< S > & *indata*, long *nRows*, long *firstRow*)

write a vector of values into a range of rows of a vector column.

see valarray version for details.

8.7.3.38 template<typename S> void CCfits::Column::write (const std::valarray< S > & *indata*, long *nRows*, long *firstRow*)

write a valarray of values into a range of rows of a vector column.

This and the equivalent vector version write a vector of values into nRows. The primary use of this is for fixed width columns, in which case Column's repeat attribute is used to determine how many elements are written to each row; if indata.size() is too small an exception will be thrown. If the column is variable width, the call will write indata.size()/nRows elements to each row.

Parameters:

indata The data to be written.

nRows the number of rows to which to write the data.

firstRow The first row to be written

8.7.3.39 template<typename S> void CCfits::Column::write (S * *indata*, long *nRows*, long *firstRow*, S * *nullValue*)

write a C array into a scalar Column, processing undefined values.

Parameters:

indata The data to be written.

nRows The size of the data array to be written

firstRow The first row to be written

nullValue Pointer to the value in the input array to be set to undefined values

8.7.3.40 template<typename S> void CCfits::Column::write (const std::valarray< S > & *indata*, long *firstRow*, S * *nullValue*)

write a valarray of values into a scalar column starting with firstRow with undefined values set to nullValue.

Parameters:

indata The data to be written.

firstRow The first row to be written

nullValue Pointer to the value in the input array to be set to undefined values

8.7.3.41 template<typename S> void CCfits::Column::write (const std::vector< S > & *indata*, long *firstRow*, S * *nullValue*)

write a vector of values into a row starting with firstRow with undefined values set to nullValue.

Parameters:

indata The data to be written.

firstRow The first row to be written

nullValue Pointer to the value in the input array to be set to undefined values

8.7.3.42 template<typename S> void CCfits::Column::write (S * *indata*, long *nRows*, long *firstRow*)

write a C array of size nRows into a scalar Column starting with row firstRow.

Parameters:

indata The data to be written.

nRows The size of the data array to be written

firstRow The first row to be written

8.7.3.43 template<typename S> void CCfits::Column::write (const std::valarray< S > & *indata*, long *firstRow*)

write a valarray of values into a scalar column starting with firstRow.

Parameters:

indata The data to be written.

firstRow The first row to be written

8.7.3.44 template<typename S> void CCfits::Column::write (const std::vector< S > & *indata*, long *firstRow*)

write a vector of values into a row starting with firstRow.

Parameters:

indata The data to be written.

firstRow The first row to be written

8.7.3.45 template<typename S> void CCfits::Column::writeArrays (const std::vector< std::valarray< S > > & *indata*, long *firstRow*, S * *nullValue*)

write a vector of valarray objects to the column, starting at row firstRow >= 1, processing undefined values.

8.7.3.46 template<typename S> void CCfits::Column::writeArrays (const std::vector< std::valarray< S > > & *indata*, long *firstRow*)

write a vector of valarray objects to the column, starting at row firstRow >= 1.

8.7.3.47 double CCfits::Column::zero () const [inline]

get TZEROn value.

TZEROn is an integer offset used in the implementation of unsigned data

The documentation for this class was generated from the following files:

- Column.h
- Column.cxx
- ColumnT.h

8.8 CCfits::FITSUtil::CVAarray Class Template Reference

function object returning C array from a vector of valarrays. see CVarray for details.

`#include <FITSUtil.h>`

Public Methods

- T* operator() (const std::vector< std::valarray< T > > &inArray)

8.8.1 Detailed Description

template<typename T> class CCfits::FITSUtil::CVAarray

function object returning C array from a vector of valarrays. see CVarray for details.

8.8.2 Member Function Documentation

8.8.2.1 template<typename T> T * CCfits::FITSUtil::CVAarray< T >::operator() (const std::vector< std::valarray< T > > & *inArray*)

operator returning C array for use with vector column data.

The documentation for this class was generated from the following file:

- FITSUtil.h

8.9 CCfits::FITSUtil::CVarray Class Template Reference

Function object class for returning C arrays from standard library objects used in the FITS library implementation.

`#include <FITSUtil.h>`

Public Methods

- T* operator() (const std::vector< T > &inArray)

8.9.1 Detailed Description

template<typename T> class CCfits::FITSUtil::CVarray

Function object class for returning C arrays from standard library objects used in the FITS library implementation.

There are 3 versions which convert std::vector<T>, std::valarray<T>, and std::vector<std::valarray<T> > objects to pointers to T, called CVarray, CAarray, and CVAarray.

An alternative function, CharArray, is provided to deal with the special case of vector string arrays.

8.9.2 Member Function Documentation

8.9.2.1 template<typename T> T * CCfits::FITSUtil::CVarray< T >::operator() (const std::vector< T > & *inArray*) [inline]

operator returning C array for use with scalar column data.

The documentation for this class was generated from the following file:

- FITSUtil.h

8.10 CCfits::ExtHDU Class Reference

base class for all FITS extension HDUs, i.e. Image Extensions and Tables.

`#include <ExtHDU.h>`

Inheritance diagram for CCfits::ExtHDU::

Public Methods

- ExtHDU (const ExtHDU &right)
- virtual ~ExtHDU ()
- virtual void readData (bool readFlag=false, const std::vector< string > &keys=std::vector< string >())=0
- const string& name () const
- virtual HDU* clone (FITSBase *p) const=0
- virtual void makeThisCurrent () const
- virtual Column& column (const string &colName) const
- virtual Column& column (int colIndex) const
- virtual long rows () const
- virtual void addColumn (ValueType type, const string &columnName, long repeatWidth, const string &colUnit=string(""), long decimals=-1, size_t columnNumber=0)
- virtual void deleteColumn (const string &columnName)
- int version () const
- void version (int value)
- std::string& name ()
- template<typename S> void write (const std::vector< long > &first, long nElements, const std::valarray< S > &data, S *nullValue)
- template<typename S> void write (long first, long nElements, const std::valarray< S > &data, S *nullValue)
- template<typename S> void write (const std::vector< long > &first, long nElements, const std::valarray< S > &data)
- template<typename S> void write (long first, long nElements, const std::valarray< S > &data)
- template<typename S> void write (const std::vector< long > &firstVertex, const std::vector< long > &lastVertex, const std::valarray< S > &data)
- template<typename S> void read (std::valarray< S > &image)
- template<typename S> void read (std::valarray< S > &image, long first, long nElements, S *nullValue)
- template<typename S> void read (std::valarray< S > &image, const std::vector< long > &first, long nElements, S *nullValue)

- template<typename S> void read (std::valarray< S > &image, const std::vector< long > &firstVertex, const std::vector< long > &lastVertex, const std::vector< long > &stride)
- template<typename S> void read (std::valarray< S > &image, long first, long nElements)
- template<typename S> void read (std::valarray< S > &image, const std::vector< long > &first, long nElements)
- template<typename S> void read (std::valarray< S > &image, const std::vector< long > &firstVertex, const std::vector< long > &lastVertex, const std::vector< long > &stride, S *nullValue)

Static Public Methods

- void readHduName (const fitsfile *fptr, int hduIndex, string &hduName, int &hduVersion)

Protected Methods

- ExtHDU (FITSBase *p=0, HduType xtype=AnyHdu, const string &hduName=string(""), int version=1)
- ExtHDU (FITSBase *p, HduType xtype, const string &hduName, int bitpix, int naxis, const std::vector< long > &axes, int version=1)
- ExtHDU (FITSBase *p, HduType xtype, int number)
- long pcount () const
- void pcount (long value)
- long gcount () const
- void gcount (long value)
- HduType xtension () const
- void xtension (HduType value)

8.10.1 Detailed Description

base class for all FITS extension HDUs, i.e. Image Extensions and Tables.

ExtHDU needs to have the combined public interface of Table objects and images. It achieves this by providing the same set of read and write operations as PHDU, and also providing the same operations for extracting columns from the extension as does Table [after which the column interface is accessible]. Differentiation between extension types operates by exception handling: .i.e. attempting to access image data structures on a Table object through the ExtHDU interface will or trying to return a Column reference from an Image extension will both throw an exception

8.10.2 Constructor & Destructor Documentation

8.10.2.1 CCfits::ExtHDU::ExtHDU (const ExtHDU & *right*)

copy constructor.

8.10.2.2 CCfits::ExtHDU::~ExtHDU () [virtual]

destructor.

8.10.2.3 CCfits::ExtHDU::ExtHDU (FITSBase * *p* = 0, HduType *xtype* = AnyHdu, const string & *hduName* = string(""), int *version* = 1) [protected]

default constructor, required as Standard Library Container content.

8.10.2.4 CCfits::ExtHDU::ExtHDU (FITSBase * *p*, HduType *xtype*, const string & *hduName*, int *bitpix*, int *naxis*, const std::vector< long > & *axes*, int *version* = 1) [protected]

writing constructor.

The writing constructor forces the user to supply a name for the HDU. The bitpix, naxes and naxis data required by this constructor are required FITS keywords for any HDUs.

8.10.2.5 CCfits::ExtHDU::ExtHDU (FITSBase * *p*, HduType *xtype*, int *number*) [protected]

ExtHDU constructor for getting ExtHDUs by number.

Necessary since EXTNAME is a reserved, not required, keyword. But a synthetic name is supplied by static ExtHDU::readHduName which is called by this constructor.

8.10.3 Member Function Documentation

8.10.3.1 void CCfits::ExtHDU::addColumn (ValueType *type*, const string & *columnName*, long *repeatWidth*, const string & *colUnit* = string(""), long *decimals* = -1, size_t *columnNumber* = 0) [virtual]

add a new column to an existing table HDU.

Parameters:

type The data type of the column to be added

columnName The name of the column to be added

repeatWidth for a string valued, this is the width of a string. For a numeric column it supplies the vector length of the rows. It is ignored for ascii table numeric data.

colUnit an optional field specifying the units of the data (TUNITn)

decimal optional parameter specifying the number of decimals for an ascii numeric column

columnNumber optional parameter specifying column number to be created. If not specified the column is added to the end. If specified, the column is inserted and the columns already read are reindexed. This parameter is provided as a convenience to support existing code rather than recommended.

Reimplemented in CCfits::AsciiTable, and CCfits::BinTable.

8.10.3.2 HDU * CCfits::ExtHDU::clone (FITSBase * *p*) const [pure virtual]

virtual copy constructor.

Reimplemented from CCfits::HDU.

Reimplemented in CCfits::AsciiTable, and CCfits::BinTable.

8.10.3.3 Column & CCfits::ExtHDU::column (int *colIndex*) const [virtual]

return a reference to a Table column specified by column index.

This version is provided for convenience; the 'return by name' version is more efficient because columns are stored in an associative array sorted by name.

Exceptions:

WrongExtensionType thrown if *this is an image extension.

Reimplemented in CCfits::Table.

8.10.3.4 Column & CCfits::ExtHDU::column (const string & *colName*) const [virtual]

return a reference to a Table column specified by name.

The overridden base class implementation ExtHDU::column throws an exception, which is thus the action to be taken if self is an image extension

Exceptions:

WrongExtensionType see above

Reimplemented in CCfits::Table.

8.10.3.5 void CCfits::ExtHDU::deleteColumn (const string & *name*) [virtual]

delete a column in a Table extension by name.

Parameters:
columnName The name of the column to be deleted.

Exceptions:
WrongExtensionType if extension is an image.

Reimplemented in CCfits::Table.

8.10.3.6 void CCfits::ExtHDU::gcount (long *value*) [inline, protected]

set required gcount keyword value.

8.10.3.7 long CCfits::ExtHDU::gcount () const [inline, protected]

return required gcount keyword value.

8.10.3.8 void makeThisCurrent () const [virtual]

move the fitsfile pointer to this current HDU.

This function should never need to be called by the user since it is called internally whenever required.

Reimplemented from CCfits::HDU.

8.10.3.9 std::string & CCfits::ExtHDU::name () [inline]

set/return the extension name.

8.10.3.10 const string & CCfits::ExtHDU::name () const [inline]

return the name of the extension.

8.10.3.11 void CCfits::ExtHDU::pcount (long *value*) [inline, protected]

set required pcount keyword value.

8.10.3.12 long CCfits::ExtHDU::pcount () const [inline, protected]

return required pcount keyword value.

8.10.3.13 template<typename S> void CCfits::ExtHDU::read (std::valarray< S > & *image*, const std::vector< long > & *firstVertex*, const std::vector< long > & *lastVertex*, const std::vector< long > & *stride*, S * *nulValue*)

read an image subset into valarray image, processing null values.

The image subset is defined by two vertices and a stride indicating the 'denseness' of the values to be picked in each dimension (a stride = (1,1,1,...) means picking every pixel in every dimension, whereas stride = (2,2,2,...) means picking every other value in each dimension.

8.10.3.14 template<typename S> void CCfits::ExtHDU::read (std::valarray< S > & *image*, const std::vector< long > & *first*, long *nElements*)

read an image section starting at a location specified by an n-tuple.

8.10.3.15 template<typename S> void CCfits::ExtHDU::read (std::valarray< S > & *image*, long *first*, long *nElements*)

read an image section starting at a specified pixel.

8.10.3.16 template<typename S> void CCfits::ExtHDU::read (std::valarray< S > & *image*, const std::vector< long > & *firstVertex*, const std::vector< long > & *lastVertex*, const std::vector< long > & *stride*)

read an image subset.

8.10.3.17 template<typename S> void CCfits::ExtHDU::read (std::valarray< S > & *image*, const std::vector< long > & *first*, long *nElements*, S * *nulValue*)

read part of an image array, processing null values.

As above except for

Parameters:

first a vector<long> representing an n-tuple giving the coordinates in the image of the first pixel.

8.10.3.18 template<typename S> void CCfits::ExtHDU::read (std::valarray< S > & *image*, long *first*, long *nElements*, S * *nulValue*)

read part of an image array, processing null values.

Implicit data conversion is supported (i.e. user does not need to know the type of the data stored. A WrongExtensionType extension is thrown if *this is not an image.

Parameters:

image The receiving container, a std::valarray reference

first The first pixel from the array to read [a long value]

nElements The number of values to read

nullValue A pointer containing the value in the table to be considered as undefined. See cfitsio for details

8.10.3.19 template<typename S> void CCfits::ExtHDU::read (std::valarray< S > & *image*)

Read image data into container.

The container image contains the entire image array after the call. This and all the other variants of read() throw a WrongExtensionType exception if called for a Table object.

8.10.3.20 void CCfits::ExtHDU::readData (bool *readFlag* = false, const std::vector< string > & *keys* = std::vector<string>()) [pure virtual]

read data from HDU depending on readFlag and keys.

Reimplemented in CCfits::AsciiTable, and CCfits::BinTable.

8.10.3.21 void CCfits::ExtHDU::readHduName (const fitsfile * *fptr*, int *hduIndex*, string & *hduName*, int & *hduVersion*) [static]

read extension name.

Used primarily to allow extensions to be specified by HDU number and provide their name for the associative array that contains them. Alternatively, if there is no name keyword in the extension, one is synthesized from the index.

8.10.3.22 long CCfits::ExtHDU::rows () const [virtual]

return the number of rows in the extension.

Exceptions:

WrongExtensionType thrown if *this is an image extension.

Reimplemented in CCfits::Table.

8.10.3.23 void CCfits::ExtHDU::version (int *value*) [inline]

set the extension version number.

8.10.3.24 int CCfits::ExtHDU::version () const [inline]

return the extension version number.

8.10.3.25 template<typename S> void CCfits::ExtHDU::write (const std::vector< long > & *firstVertex*, const std::vector< long > & *lastVertex*, const std::valarray< S > & *data*)

write a subset (generalize slice) of data to the image.

A generalized slice/subset is a subset of the image (e.g. one plane of a data cube of size <= the dimension of the cube). It is specified by two opposite vertices. The equivalent cfitsio call does not support undefined data processing so there is no version that allows a null value to be specified.

Parameters:

firstVertex the coordinates specifying lower and upper vertices of the n-dimensional slice

lastVertex

data The data to be written

8.10.3.26 template<typename S> void CCfits::ExtHDU::write (long *first*, long *nElements*, const std::valarray< S > & *data*)

write array starting from specified pixel number, without undefined data processing.

8.10.3.27 template<typename S> void CCfits::ExtHDU::write (const std::vector< long > & *first*, long *nElements*, const std::valarray< S > & *data*)

write array starting from specified n-tuple, without undefined data processing.

8.10.3.28 template<typename S> void CCfits::ExtHDU::write (long *first*, long *nElements*, const std::valarray< S > & *data*, S * *nulValue*)

write array to image starting with a specified pixel and allowing undefined data to be processed.

parameters after the first are as for version with n-tuple specifying first element. these two version are equivalent, except that it is possible for the first pixel number to exceed the range of 32-bit integers, which is how long datatype is commonly implemented.

8.10.3.29 template<typename S> void CCfits::ExtHDU::write (const std::vector< long > & *first*, long *nElements*, const std::valarray< S > & *data*, S * *nulValue*)

Write a set of pixels to an image extension with the first pixel specified by an n-tuple, processing undefined data.

All the overloaded versions of ExtHDU::write perform operations on ∗this if it is an image and throw a WrongExtensionType exception if not. Where appropriate, alternate versions allow undefined data to be processed

Parameters:

first an n-tuple of dimension equal to the image dimension specifying the first pixel in the range to be written

nElements number of pixels to be written

data array of data to be written

pointer to null value (data with this value written as undefined; needs the BLANK keyword to have been specified).

8.10.3.30 void CCfits::ExtHDU::xtension (HduType *value*) [inline, protected]

set the extension type.

8.10.3.31 HduType CCfits::ExtHDU::xtension () const [inline, protected]

return the extension type.

allowed values are ImageHDU, AsciiTbl, and BinaryTbl

The documentation for this class was generated from the following files:

- ExtHDU.h
- ExtHDU.cxx
- ExtHDUT.h
- FITS.h

8.11 CCfits::FITS Class Reference

Memory object representation of a disk FITS file.

`#include <FITS.h>`

Public Methods

- FITS (const string &name, RWmode mode=Read, bool readDataFlag=false, const std::vector< string > &primaryKeys=std::vector< string >())

- FITS (const string &name, RWmode mode, const string &hduName, bool readDataFlag=false, const std::vector< string > &hduKeys=std::vector< string >(), const std::vector< string > &primaryKey=std::vector< string >(), int version=1)
- FITS (const string &name, RWmode mode, const std::vector< string > &hduNames, bool readDataFlag=false, const std::vector< string > &primaryKey=std::vector< string >())
- FITS (const string &fileName, const FITS &source)
- FITS (const string &name, RWmode mode, const std::vector< string > &hduNames, const std::vector< std::vector< string > > &hduKeys, bool readDataFlag=false, const std::vector< string > &primaryKeys=std::vector< string >(), const std::vector< int > &hduVersions=std::vector< int >())
- FITS (const string &name, int bitpix, int naxis, long *naxes)
- FITS (const string &name, RWmode mode, int hduIndex, bool readDataFlag=false, const std::vector< string > &hduKeys=std::vector< string >(), const std::vector< string > &primaryKey=std::vector< string >())
- FITS (const string &name, RWmode mode, const std::vector< string > &searchKeys, const std::vector< string > &searchValues, bool readDataFlag=false, const std::vector< string > &hduKeys=std::vector< string >(), const std::vector< string > &primaryKey=std::vector< string >(), int version=1)
- ~FITS ()
- void deleteExtension (const std::string &doomed, int version=1)
- void read (const string &hduName, bool readDataFlag=false, const std::vector< string > &keys=std::vector< string >(), int version=0)
- void read (const std::vector< string > &hduNames, bool readDataFlag=false)
- void read (const std::vector< string > &hduNames, const std::vector< std::vector< string > > &keys, bool readDataFlag=false, const std::vector< int > &hduVersions=std::vector< int >())
- FITS* clone () const
- const ExtHDU& extension (int i) const
- ExtHDU& extension (int i)
- const ExtHDU& extension (const string &hduName, int version=1) const
- const PHDU& pHDU () const
- PHDU& pHDU ()
- ExtHDU& extension (const string &hduName, int version=1)
- void read (int hduIndex, bool readDataFlag=false, const std::vector< string > &keys=std::vector< string >())
- Table* addTable (string &hduName, int rows, const std::vector< string > &columnName=std::vector< string >(), const std::vector< string > &columnFmt=std::vector< string >(), const std::vector< string > &columnUnit=std::vector< string >(), HduType type=BinaryTbl, int version=1)
- ExtHDU* addImage (const string &hduName, int bpix, std::vector< long > &naxes, int version=1)

- void destroy () throw ()
- void flush ()
- const string& currentExtensionName () const
- void read (const std::vector< string > &searchKeys, const std::vector< string > &searchValues, bool readDataFlag=false, const std::vector< string > &hduKeys=std::vector< string >(), int version=1)
- const std::multimap<string,ExtHDU*>& extension () const
- void resetPosition ()
- const string& name () const
- void copy (const HDU &source)
- Table& filter (const string &expression, ExtHDU &inputTable, bool overwrite=true, bool readData=false)
- ExtHDU& currentExtension ()
- void deleteExtension (int doomed)

Static Public Methods

- void clearErrors ()
- bool verboseMode ()
- void setVerboseMode (bool value)

8.11.1 Detailed Description

Memory object representation of a disk FITS file.

Constructors are provided to get FITS data from an existing file or to create new FITS data sets. Overloaded versions allow the user to

a) read from one or more specified extensions, specified by EXTNAME and VERSION or by HDU number. b either just header information or data on construction c) to specify scalar keyword values to be read on construction d) to open and read an extension that has specified keyword values e) create a new FITS object and corresponding file, including an empty primary header.

The memory fits object as constructed is always an image of a valid FITS object, i.e. a primary HDU is created on construction.

calling the destructor closes the disk file, so that FITS files are automatically deleted at the end of scope unless other arrangements are made.

8.11.2 Constructor & Destructor Documentation

8.11.2.1 CCfits::FITS::FITS (const string & *name*, RWmode *mode* = Read, bool *readDataFlag* = false, const std::vector< string > & *primaryKeys* = std::vector<string>())

basic constructor.

This basic constructor makes a FITS object from the given filename. The filename string is passed directly to the cfitsio library: thus all of the extended filename syntax described in the cfitsio manual should work as documented.

If the mode is Read [default], it will read all of the headers in the file, and all of the data if the readDataFlag is supplied as true. It will also read optional primary keys.

The file name is the only required argument. If the mode is Write and the file does not already exist, a default primary HDU will be created in the file with BITPIX=8 and NAXIS=0: this mode is designed for writing FITS files with table extensions only. For files with image data the constructor that specified the data type and number of axes should be called.

Parameters:

name The name of the FITS file to be read/written

mode The read/write mode: must be Read or Write

readDataFlag boolean: read data on construction if true

primaryKeys Allows optional reading of primary header keys on construction

Exceptions:

NoSuchHDU thrown on HDU seek error either by index or {name,version}

FitsError thrown on non-zero status code from cfitsio when not overriden by FitsException error to produce more illuminating message.

8.11.2.2 CCfits::FITS::FITS (const string & *name*, RWmode *mode*, const string & *hduName*, bool *readDataFlag* = false, const std::vector< string > & *hduKeys* = std::vector<string>(), const std::vector< string > & *primaryKey* = std::vector<string>(), int *version* = 1)

Open a FITS file and read a single specified HDU.

This and similar constructor variants support reading table data.

Optional arguments allow the reading of primary header keys and specified data from hduName, the HDU to be read. An object representing the primary HDU is always created: if it contains an image, that image may be read by subsequent calls.

Parameters:

name The name of the FITS file to be read

mode The read/write mode: takes values Read or Write

hduName The name of the HDU to be read.

hduKeys Optional array of keywords to be read from the HDU

version Optional version number. If not supplied the first HDU with name *hduName* is read see above for other parameter definitions

8.11.2.3 CCfits::FITS::FITS (const string & *name*, RWmode *mode*, const std::vector< string > & *hduNames*, bool *readDataFlag* = false, const std::vector< string > & *primaryKey* = std::vector<string>())

This is intended as a convenience where the file consists of single versions of HDUs and data only, not keys are to be read.

Parameters:

hduNames array of HDU names to be read. see above for other parameter definitions.

8.11.2.4 CCfits::FITS::FITS (const string & *fileName*, const FITS & *source*)

create a new FITS object and corresponding file with copy of the primary header of the source.

Parameters:

fileName New file to be created.

source A previously created FITS object to be copied.

see above for other parameter definitions.

8.11.2.5 CCfits::FITS::FITS (const string & *name*, RWmode *mode*, const std::vector< string > & *hduNames*, const std::vector< std::vector< string > > & *hduKeys*, bool *readDataFlag* = false, const std::vector< string > & *primaryKeys* = std::vector<string>(), const std::vector< int > & *hduVersions* = std::vector<int>())

FITS read constructor in full generality.

Parameters:

hduVersions an optional version number for each HDU to be read

hduKeys an array of keywords for each HDU to be read. see above for other parameter definitions.

8.11.2.6 CCfits::FITS::FITS (const string & *name*, int *bitpix*, int *naxis*, long * *naxes*)

Constructor for creating new FITS objects containing images.

This constructor is only called for creating new files (mode is not an argument) and creates a new primary HDU with the datatype & axes specified by bitpix, naxis, and naxes. The data are added to the new fits object and file by subsequent calls to FITS::pHDU().write(<arguments>)

Valid values of bitpix are given in the fitsio.h header. If the filename corresponds to an existing file and does not start with the '!' character the construction will fail with a CantCreate exception.

The arguments are:

Parameters:

name The file to be written to disk

bitpix the datatype of the primary image (see cfitsio documentation for allowed values.

naxis the data dimension of the primary image

naxes the array of axis lengths for the primary image. Ignored if naxis =0, i.e. the primary header is empty. extensions can be added arbitrarily to the file after this constructor is called. The constructors should write header information to disk:

8.11.2.7 CCfits::FITS::FITS (const string & *name*, RWmode *mode*, int *hduIndex*, bool *readDataFlag* = false, const std::vector< string > & *hduKeys* = std::vector<string>(), const std::vector< string > & *primaryKey* = std::vector<string>())

read a single numbered HDU.

Constructor analogous to the version that reads by name. This is required since HDU extensions are not required to have the EXTNAME or HDUNAME keyword by the standard. If there is no name, a dummy name based on the HDU number is created and becomes the key.

Parameters:

hduIndex The index of the HDU to be read. see above for other parameter definitions.

8.11.2.8 CCfits::FITS::FITS (const string & *name*, RWmode *mode*, const std::vector< string > & *searchKeys*, const std::vector< string > & *searchValues*, bool *readDataFlag* = false, const std::vector< string > & *hduKeys* = std::vector<string>(), const std::vector< string > & *primaryKey* = std::vector<string>(), int *version* = 1)

open fits file and read HDU which contains supplied keywords with [optional] specified values (sometimes one just wants to know that the keyword is present).

Optional parameters allows the reading of specified primary HDU keys and specified columns and keywords in the HDU of interest.

Parameters:

name The name of the FITS file to be read

mode The read/write mode: must be Read or Write

searchKeys A string vector of keywords to search for in each header

searchValues A string vector of values those keywords are required to have for success. Note that the keys must be of type string. If any value does not need to be checked the corresponding searchValue element can be empty.

readDataFlag boolean: if true, read data if HDU is found

hduKeys Allows optional reading of keys in the HDU that is searched for if it is successfully found

primaryKeys Allows optional reading of primary header keys on construction

version Optional version number. If specified, checks the EXTVERS keyword.

Exceptions:

FitsError thrown on non-zero status code from cfitsio when not overriden by FitsException error to produce more illuminating message.

8.11.2.9 CCfits::FITS::~FITS ()

destructor.

8.11.3 Member Function Documentation

8.11.3.1 ExtHDU * CCfits::FITS::addImage (const string & *hduName*, int *bpix*, std::vector< long > & *naxes*, int *version* = 1)

Add an image extension to an existing FITS object. (File with w or rw access).

Does not make primary images, which are built in the constructor for the FITS file. The image data is not added here: it can be added by a second call.

Todo:

Add a function for replacing the primary image

8.11.3.2 Table * CCfits::FITS::addTable (string & *hduName*, int *rows*, const std::vector< string > & *columnName* = std::vector<string>(), const std::vector< string > & *columnFmt* = std::vector<string>(), const std::vector< string > & *columnUnit* = std::vector<string>(), HduType *type* = BinaryTbl, int *version* = 1)

Add a table extension to an existing FITS object. Add extension to FITS object for file with w or rw access.

Parameters:

rows The number of rows in the table to be created.

columnName A vector containing the table column names

columnFmt A vector containing the table column formats

columnUnit (Optional) a vector giving the units of the columns.

HduType The table type - AsciiTbl or BinaryTbl (defaults to BinaryTbl) the lists of columns are optional - one can create an empty table extension but if supplied, colType, columnName and colFmt must have equal dimensions.

Todo:

the code should one day check that the version keyword is higher than any other versions already added to the FITS object (although cfitsio doesn't do this either).

8.11.3.3 void CCfits::FITS::clearErrors () [static]

clear the error stack and set status to zero.

8.11.3.4 FITS * CCfits::FITS::clone () const

clone function: to allow implementation later of reference counting.

8.11.3.5 void CCfits::FITS::copy (const HDU & *source*)

copy the HDU source into the FITS object.

This function adds a copy of an HDU from another file into *this. It does not create a duplicate of an HDU in the file associated with *this.

8.11.3.6 ExtHDU & CCfits::FITS::currentExtension ()

return a non-const reference to whichever is the current extension.

8.11.3.7 const string & CCfits::FITS::currentExtensionName () const

return the name of the extension that the fitsfile is currently addressing.

If the extension in question does not have an EXTNAME or HDUNAME keyword, then the function returns HDUn, where n is the sequential HDU index number (primary HDU = 0).

8.11.3.8 void CCfits::FITS::deleteExtension (int *doomed*)

Delete extension specified by extension number.

\overload

8.11.3.9 void CCfits::FITS::deleteExtension (const std::string & *doomed*, int *version* = 1)

Delete extension specified by name and version number.

Removes extension from FITS object and memory copy.

Parameters:

doomed the name of the extension to be deleted

version an optional version number, the EXTVER keyword, defaults to 1

Exceptions:

NoSuchHDU Thrown if there is no extension with the specified version number

FitsError Thrown if there is a non-zero status code from cfitsio, e.g. if the delete operation is applied to a FITS file opened for read only access.

8.11.3.10 void CCfits::FITS::destroy () throw ()

Erase FITS object and close corresponding file.

Force deallocation and erase of elements of a FITS memory object. Allows a reset of everything inside the FITS object, and closes the file. The object is inaccessible after this call.

destroy is public to allow users to reuse a symbol for a new file, but it is identical in operation to the destructor.

8.11.3.11 const std::multimap< string, ExtHDU *> & CCfits::FITS::extension<string,ExtHDU*> () const

return const reference to the extension container.

This is useful for such operations as extension().size() etc.

8.11.3.12 ExtHDU & CCfits::FITS::extension (const string & *hduName*, int *version* = 1) [inline]

return FITS extension by name and (optionally) version number.

8.11.3.13 const ExtHDU & CCfits::FITS::extension (const string & *hduName*, int *version* = 1) const

return FITS extension by name and (optionally) version number.

8.11.3.14 ExtHDU & CCfits::FITS::extension (int *i*)

return FITS extension by index number. non-const version. see const version for details.

8.11.3.15 const ExtHDU & CCfits::FITS::extension (int *i*) const

return FITS extension by index number. N.B. The input index number is currently defined as enumerating extensions, so the extension(1) returns HDU number 2.

8.11.3.16 Table & CCfits::FITS::filter (const string & *expression*, ExtHDU & *inputTable*, bool *overwrite* = true, bool *readData* = false)

Filter the rows of the inputTable with the condition expression, and return a reference to the resulting Table.

This function provides an object oriented version of cfitsio's fits_select_rows call. The expression string is any boolean expression involving the names of the columns in the input table (e.g., if there were a column called "density", a valid expression might be "DENSITY > 3.5": see the cfitsio documentation for further details).

[N.B. the "append" functionality described below does not work when linked with cfitsio 2.202 or prior because of a known issue with that version of the library. This causes the output to be a new extension with a correct header copy and version number but without the filtered data]. If the inputTable is an Extension HDU of this FITS object, then if overwrite is true the operation will overwrite the inputTable with the filtered version, otherwise it will append a new HDU with the same extension name but the next highest version (EXTVER) number available.

8.11.3.17 void CCfits::FITS::flush ()

flush buffer contents to disk.

Provides manual control of disk writing operation. Image data are flushed automatically to disk after the write operation is completed, but not column data.

8.11.3.18 const string & CCfits::FITS::name () const

return filename of file corresponding to FITS object.

8.11.3.19 PHDU & CCfits::FITS::pHDU ()

return a reference to the primary HDU.

8.11.3.20 const PHDU & CCfits::FITS::pHDU () const

return a const reference to the primary HDU.

8.11.3.21 void CCfits::FITS::read (const std::vector< string > & *searchKeys*, const std::vector< string > & *searchValues*, bool *readDataFlag* = false, const std::vector< string > & *hduKeys* = std::vector<string>(), int *version* = 1)

read method for read header or HDU that contains specified keywords.

Parameters:

searchKeys A string vector of keywords to search for in each header

searchValues A string vector of values those keywords are required to have for success. Note that the keys must be of type string. If any value does not need to be checked the corresponding searchValue element can be empty.

readDataFlag boolean: if true, read data if HDU is found

hduKeys Allows optional reading of keys in the HDU that is searched for if it is successfully found

primaryKeys Allows optional reading of primary header keys on construction

version Optional version number. If specified, checks the EXTVERS keyword.

8.11.3.22 void CCfits::FITS::read (int *hduIndex*, bool *readDataFlag* = false, const std::vector< string > & *keys* = std::vector<string>())

read an HDU specified by index number.

This is provided to allow reading of HDUs after construction. see above for parameter definitions.

8.11.3.23 void CCfits::FITS::read (const std::vector< string > & *hduNames*, const std::vector< std::vector< string > > & *keys*, bool *readDataFlag* = false, const std::vector< int > & *hduVersions* = std::vector<int>())

get data from a set of HDUs from disk file, specifying keys and version numbers.

This is provided to allow reading of HDUs after construction. see above for parameter definitions.

8.11.3.24 void CCfits::FITS::read (const std::vector< string > & *hduNames*, bool *readDataFlag* = false)

get data from a set of HDUs from disk file.

This is provided to allow reading of HDUs after construction. see above for parameter definitions.

8.11.3.25 void CCfits::FITS::read (const string & *hduName*, bool *readDataFlag* = false, const std::vector< string > & *keys* = std::vector<string>(), int *version* = 0)

get data from single HDU from disk file.

This is provided to allow the adding of additional HDUs to the FITS object after construction of the FITS object. After the read() functions have been called for the FITS object, subsequent read method to the Primary, ExtHDU, and Column objects will retrieve data from the FITS object in memory (those methods can be called to read data in those HDU objects that was not read when the HDU objects were constructed.

All the read functions will throw NoSuchHDU exceptions on seek errors since they involve constructing HDU objects.

The parameter definitions are as documented for the corresponding constructor.

8.11.3.26 void CCfits::FITS::resetPosition ()

explicit call to set the fits file pointer to the primary.

8.11.3.27 void CCfits::FITS::setVerboseMode (bool *value*) [inline, static]

set verbose setting for library.

8.11.3.28 bool CCfits::FITS::verboseMode () [inline, static]

return verbose setting for library.

If true, all messages that are reported by exceptions are printed to std::cerr.

The documentation for this class was generated from the following files:

- FITS.h
- FITS.cxx

8.12 CCfits::FitsError Class Reference

FitsError is the exception thrown by non-zero cfitsio status codes.

`#include <FitsError.h>`

Inheritance diagram for CCfits::FitsError::

Public Methods

- FitsError (int errornum, bool silent=true)

8.12.1 Detailed Description

FitsError is the exception thrown by non-zero cfitsio status codes.

8.12.2 Constructor & Destructor Documentation

8.12.2.1 CCfits::FitsError::FitsError (int *errornum*, bool *silent* = true)

ctor for cfitsio exception: translates status code into cfitsio error message.

The exception prefixes the string "Fits Error: " to the message printed by cfitsio.

Parameters:

errornum The cfitsio status code produced by the error.

silent A boolean controlling the printing of messages

The documentation for this class was generated from the following files:

- FitsError.h
- FitsError.cxx

8.13 CCfits::FitsException Class Reference

FitsException is the base class for all exceptions thrown by this library.

`#include <FitsError.h>`

Inheritance diagram for CCfits::FitsException::

Public Methods

- FitsException (const string &msg, bool &silent)

8.13.1 Detailed Description

FitsException is the base class for all exceptions thrown by this library.

All exceptions derived from this class can be caught by a single 'catch' clause catching FitsException by reference (which is the point of this base class design).

A static "verboseMode" parameter is provided by the FITS class to control diagnostics - if FITS::verboseMode() is true, all diagnostics are printed (for debugging purposes). If not, then a boolean *silent* determines printing of messages. Each exception derived from FitsException must define a default value for the *silent* parameter.

8.13.2 Constructor & Destructor Documentation

8.13.2.1 CCfits::FitsException::FitsException (const string & *msg*, bool & *silent*)

Parameters:

diag A diagnostic string to be printed optionally.

silent A boolean controlling the printing of messages

The documentation for this class was generated from the following files:

- FitsError.h
- FitsError.cxx

8.14 CCfits::FitsFatal Class Reference

[potential] base class for exceptions to be thrown on internal library error.

`#include <FitsError.h>`

Public Methods

- FitsFatal (const string &diag)

8.14.1 Detailed Description

[potential] base class for exceptions to be thrown on internal library error.

As of this version there are no subclasses. This error requests that the user reports this circumstance to HEASARC.

8.14.2 Constructor & Destructor Documentation

8.14.2.1 CCfits::FitsFatal::FitsFatal (const string & *diag*)

Prints a message starting "∗∗∗ CCfits Fatal Error: ..." and calls *terminate()*.

Parameters:

diag A diagnostic string to be printed identifying the context of the error.

The documentation for this class was generated from the following files:

- FitsError.h
- FitsError.cxx

8.15 CCfits::HDU Class Reference

Base class for all HDU [Header-Data Unit] objects.

`#include <HDU.h>`

Inheritance diagram for CCfits::HDU::

Public Methods

- HDU (const HDU &right)
- bool operator== (const HDU &right) const
- bool operator!= (const HDU &right) const
- virtual HDU* clone (FITSBase *p) const=0
- fitsfile* fitsPointer () const
- virtual void makeThisCurrent () const
- std::map<string, Keyword*> keyWord ()
- Keyword& keyWord (const string &keyName)
- const string& getHistory ()
- const string& getComments ()
- void writeHistory (const string &history="Generic History String")
- void writeComment (const string &comment="Generic Comment")
- void writeDate ()
- FITSBase* parent () const
- void readAllKeys ()
- long axes () const
- long axis (size_t index) const
- void index (int value)
- void deleteKey (const std::string &doomed)
- long bitpix () const
- void bitpix (long value)
- int index () const
- const string& history () const
- const string& comment () const
- virtual double zero () const
- virtual double scale () const
- const std::map<string,Keyword*>& keyWord () const
- const Keyword& keyWord (const string &keyname) const
- void setKeyWord (const string &keyname, Keyword &value)
- template<typename T> Keyword& addKey (const string &name, T val, const string &comment)
- template<typename T> void readKey (const string &keyName, T &val)
- template<typename T> void readKeys (std::vector< string > &keyNames, std::vector< T > &vals)

Protected Methods

- HDU (FITSBase *p=0)
- HDU (FITSBase *p, int bitpix, int naxis, const std::vector< long > &axes)
- virtual ~HDU ()
- std::vector<long>& naxes ()

8.15.1 Detailed Description

Base class for all HDU [Header-Data Unit] objects.

HDU objects in CCfits are either PHDU (Primary HDU objects) or ExtHDU (Extension HDU) objects. Following the behavior. ExtHDUs are further subclassed into ImageExt or Table objects, which are finally AsciiTable or BinTable objects.

HDU's public interface gives access to properties that are common to all HDUs, largely required keywords, and functions that are common to all HDUs, principally the manipulation of keywords and their values.

HDUs must be constructed by HDUCreator objects which are called by FITS methods. Each HDU has an embedded pointer to a FITSBase object, which is private to FITS [FITSBase is a pointer encapsulating the resources of FITS. For details of this coding idiom see Exceptional C++ by Herb Sutter (2000) and references therein].

8.15.2 Constructor & Destructor Documentation

8.15.2.1 CCfits::HDU::HDU (const HDU & *right*)

copy constructor.

8.15.2.2 CCfits::HDU::HDU (FITSBase * *p* = 0) [protected]

default constructor, called by HDU subclasses that read from FITS files.

8.15.2.3 CCfits::HDU::HDU (FITSBase * *p*, int *bitpix*, int *naxis*, const std::vector< long > & *axes*) [protected]

constructor for creating new HDU objects, called by HDU subclasses writing to FITS files.

8.15.2.4 CCfits::HDU::~HDU () [protected, virtual]

destructor.

8.15.3 Member Function Documentation

8.15.3.1 template<typename T> Keyword & CCfits::HDU::addKey (const string & *name*, T *value*, const string & *comment*)

create a new keyword in the HDU with specified value and comment fields.

the function returns a reference to keyword object just created.

Parameters:

Parameters:

name (std::string) The keyword name

value (T = std::string, float, std::complex<float>, int, or bool

comment (std::string) the keyword value

Note the limitations on T, which arise for historical reasons (i.e. since keywords do not contain explicit type information, routines in CCfits must assign a type). The limitation may be removed in a future release

It is possible however to create a keyword of any of the allowed data types in fitsio (see the cfitsio manual section 4.3). The required calls are, e.g. for a floating point double valued keyword:

NewKeyword<double> keyCreate(hdu, double val1); Keyword* key1 = Keyword keyCreate.MakeKeyword(keyName1, const string& comment); keyCreate.keyData(val2); Keyword* key2 = Keyword keyCreate.MakeKeyword(keyName2, const string& comment);

key1->write(); key2->write();

... which is essentially what addKey does for its allowed types.

8.15.3.2 long CCfits::HDU::axes () const [inline]

return the number of axes in the HDU data section (always 2 for tables).

8.15.3.3 long CCfits::HDU::axis (size_t *index*) const [inline]

return the length of HDU data axis i.

8.15.3.4 void CCfits::HDU::bitpix (long *value*) [inline]

set the data type keyword.

8.15.3.5 long CCfits::HDU::bitpix () const [inline]

return the data type keyword.

Takes values denoting the image data type for images, and takes the fixed value 8 for tables.

8.15.3.6 HDU * CCfits::HDU::clone (FITSBase * *p*) const [pure virtual]

virtual copy constructor, to be implemented in subclasses.

Reimplemented in CCfits::AsciiTable, CCfits::BinTable, CCfits::ExtHDU, and CCfits::PHDU.

8.15.3.7 const string & CCfits::HDU::comment () const [inline]

return the comment string previously read by getComment().

8.15.3.8 void CCfits::HDU::deleteKey (const std::string & *doomed*)

delete a keyword from the header.

removes *doomed*from the FITS file and from the FITS object

8.15.3.9 fitsfile * CCfits::HDU::fitsPointer () const

return the fitsfile pointer for the FITS object containing the HDU.

8.15.3.10 const string & CCfits::HDU::getComments ()

read the comments from the HDU and add it to the FITS object.

The comment string found in the header is concatenated and returned to the calling function

8.15.3.11 const string & CCfits::HDU::getHistory ()

read the history information from the HDU and add it to the FITS object.

The history string found in the header is concatenated and returned to the calling function

8.15.3.12 const string & CCfits::HDU::history () const [inline]

return the history string previously read by getHistory().

8.15.3.13 int CCfits::HDU::index () const [inline]

return the HDU number.

8.15.3.14 void CCfits::HDU::index (int *value*) [inline]

set the HDU number.

8.15.3.15 const Keyword & CCfits::HDU::keyWord (const string & *keyname*) const [inline]

return a (previously read) keyword from the HDU object. const version.

8.15.3.16 const std::map< string, Keyword *> & CCfits::HDU::keyWord () const [inline]

return the associative array containing the HDU Keywords that have been read so far.

8.15.3.17 Keyword & CCfits::HDU::keyWord (const string & *keyName*) [inline]

return a (previously read) keyword from the HDU object.

8.15.3.18 std::map< string, Keyword *> CCfits::HDU::keyWord () [inline]

return the associative array containing the HDU keywords so far read.

8.15.3.19 void CCfits::HDU::makeThisCurrent () const [virtual]

move the fitsfile pointer to this current HDU.

This function should never need to be called by the user since it is called internally whenever required.

Reimplemented in CCfits::ExtHDU.

8.15.3.20 std::vector< long > & CCfits::HDU::naxes () [inline, protected]

return the HDU data axis array.

8.15.3.21 bool CCfits::HDU::operator!= (const HDU & *right*) const

inequality operator.

8.15.3.22 bool CCfits::HDU::operator== (const HDU & *right*) const

equality operator.

8.15.3.23 FITSBase * CCfits::HDU::parent () const

return reference to the pointer representing the FITSBase object containing the HDU.

8.15.3.24 void CCfits::HDU::readAllKeys ()

read all of the keys in the header.

This member function reads keys that are not meta data for columns or image information, [which are considered to be part of the column or image objects]. Also, history and comment keys are read and returned by getHistory() and getComment().

Note that readAllKeys can only construct keys of type string, float, complex<float>, integer, and bool because the FITS header records do not encode exact type information.

8.15.3.25 template<typename T> void CCfits::HDU::readKey (const string & *keyName*, T & *val*)

read a keyword of specified type from the header of a disk FITS file and return its value.

T is one of the types std::string, float, int, std::complex<float>, and bool.

8.15.3.26 template<typename T> void CCfits::HDU::readKeys (std::vector< string > & *keyNames*, std::vector< T > & *vals*)

read a set of specified keywords of the same data type from the header of a disk FITS file and return their values.

T is one of the types std::string, float, int, std::complex<float>, and bool.

8.15.3.27 double CCfits::HDU::scale () const [inline, virtual]

return the BSCALE keyword value.

Reimplemented in CCfits::PHDU.

8.15.3.28 void CCfits::HDU::setKeyWord (const string & *keyname*, Keyword & *value*) [inline]

set Keyword keyname to specified value.

8.15.3.29 void CCfits::HDU::writeComment (const string & *comment* = "Generic Comment")

write a comment string.

A default value for the string is given ("Generic Comment String") so users can put a placeholder call to this function in their code.

8.15.3.30 void CCfits::HDU::writeDate ()

write a date string to ∗this.

8.15.3.31 void CCfits::HDU::writeHistory (const string & *history* = "Generic History String")

write a history string.

A default value for the string is given ("Generic History String") so users can put a placeholder call to this function in their code.

8.15.3.32 double CCfits::HDU::zero () const [inline, virtual]

return the BZERO keyword value.

Reimplemented in CCfits::PHDU.

The documentation for this class was generated from the following files:

- HDU.h
- FITS.h
- HDU.cxx

8.16 CCfits::Column::InsufficientElements Class Reference

Exception thrown if the data supplied for a write operation is less than declared.

`#include <Column.h>`

Inheritance diagram for CCfits::Column::InsufficientElements::

Public Methods

- InsufficientElements (const string &msg, bool silent=true)

8.16.1 Detailed Description

Exception thrown if the data supplied for a write operation is less than declared.

This circumstance generates an exception to avoid unexpected behaviour after the write operation is completed. It can be avoided by resizing the input array appropriately.

8.16.2 Constructor & Destructor Documentation

8.16.2.1 CCfits::Column::InsufficientElements::InsufficientElements (const string & *msg*, bool *silent* = true)

Exception ctor, prefixes the string "FitsError: not enough elements supplied for write operation: " before the specific message.

Parameters:

diag A specific diagnostic message, usually the column name

silent if true, print message whether FITS::verboseMode is set or not.

The documentation for this class was generated from the following files:

- Column.h
- Column.cxx

8.17 CCfits::Column::InvalidDataType Class Reference

Exception thrown for invalid data type inputs.

`#include <Column.h>`

Inheritance diagram for CCfits::Column::InvalidDataType::

Public Methods

- InvalidDataType (const string &str="", bool silent=true)

8.17.1 Detailed Description

Exception thrown for invalid data type inputs.

This exception is thrown if the user requests an implicit data type conversion to a datatype that is not one of the supported types (see fitsio.h for details).

8.17.2 Constructor & Destructor Documentation

8.17.2.1 CCfits::Column::InvalidDataType::InvalidDataType (const string & *str* = "", bool *silent* = true)

Exception ctor, prefixes the string "FitsError: Incorrect data type: " before the specific message.

Parameters:

str A specific diagnostic message

silent if true, print message whether FITS::verboseMode is set or not.

The documentation for this class was generated from the following files:

- Column.h
- Column.cxx

8.18 CCfits::HDU::InvalidExtensionType Class Reference

exception to be thrown if user requests extension type that can not be understood as ImageExt, AsciiTable or BinTable.

`#include <HDU.h>`

Inheritance diagram for CCfits::HDU::InvalidExtensionType::

Public Methods

- InvalidExtensionType (const string &diag, bool silent=true)

8.18.1 Detailed Description

exception to be thrown if user requests extension type that can not be understood as ImageExt, AsciiTable or BinTable.

8.18.2 Constructor & Destructor Documentation

8.18.2.1 CCfits::HDU::InvalidExtensionType::InvalidExtensionType (const string & *diag*, bool *silent* = true)

Exception ctor, prefixes the string "Fits Error: Extension Type: " before the specific message.

Parameters:

diag A specific diagnostic message

silent if true, print message whether FITS::verboseMode is set or not.

The documentation for this class was generated from the following files:

- HDU.h
- HDU.cxx

8.19 CCfits::HDU::InvalidImageDataType Class Reference

exception to be thrown if user requests creation of an image of type not supported by cfitsio.

`#include <HDU.h>`

Inheritance diagram for CCfits::HDU::InvalidImageDataType::

Public Methods

- InvalidImageDataType (const string &diag, bool silent=true)

8.19.1 Detailed Description

exception to be thrown if user requests creation of an image of type not supported by cfitsio.

8.19.2 Constructor & Destructor Documentation

8.19.2.1 CCfits::HDU::InvalidImageDataType::InvalidImageDataType (const string & *diag*, bool *silent* = true)

Exception ctor, prefixes the string "Fits Error: Invalid Data Type for Image " before the specific message.

Parameters:

diag A specific diagnostic message

silent if true, print message whether FITS::verboseMode is set or not.

The documentation for this class was generated from the following files:

- HDU.h
- HDU.cxx

8.20 CCfits::Column::InvalidNumberOfRows Class Reference

Exception thrown if user enters a non-positive number for the number of rows to write.

`#include <Column.h>`

Inheritance diagram for CCfits::Column::InvalidNumberOfRows::

Public Methods

- InvalidNumberOfRows (size_t number, bool silent=true)

8.20.1 Detailed Description

Exception thrown if user enters a non-positive number for the number of rows to write.

8.20.2 Constructor & Destructor Documentation

8.20.2.1 CCfits::Column::InvalidNumberOfRows::InvalidNumberOfRows (size_t *number*, bool *silent* = true)

Exception ctor, prefixes the string "Fits Error: number of rows to write must be positive " before the specific message.

Parameters:

number The number of rows entered.

silent if true, print message whether FITS::verboseMode is set or not.

The documentation for this class was generated from the following files:

- Column.h
- Column.cxx

8.21 CCfits::Column::InvalidRowNumber Class Reference

Exception thrown on attempting to read a row number beyond the end of a table.

`#include <Column.h>`

Inheritance diagram for CCfits::Column::InvalidRowNumber::

Public Methods

- InvalidRowNumber (const string &diag, bool silent=true)

8.21.1 Detailed Description

Exception thrown on attempting to read a row number beyond the end of a table.

8.21.2 Constructor & Destructor Documentation

8.21.2.1 CCfits::Column::InvalidRowNumber::InvalidRowNumber (const string & *diag*, bool *silent* = true)

Exception ctor, prefixes the string "FitsError: Invalid Row Number - Column: " before the specific message.

Parameters:

diag A specific diagnostic message, usually the column name.

silent if true, print message whether FITS::verboseMode is set or not.

The documentation for this class was generated from the following files:

- Column.h
- Column.cxx

8.22 CCfits::Column::InvalidRowParameter Class Reference

Exception thrown on incorrect row writing request.

`#include <Column.h>`

Inheritance diagram for CCfits::Column::InvalidRowParameter::

Public Methods

- InvalidRowParameter (const string &diag, bool silent=true)

8.22.1 Detailed Description

Exception thrown on incorrect row writing request.

This exception is thrown if the user requests writing more data than a fixed width row can accommodate. An exception is thrown rather than a truncation because it is likely that the user will not otherwise realize that data loss is happening.

8.22.2 Constructor & Destructor Documentation

8.22.2.1 CCfits::Column::InvalidRowParameter::InvalidRowParameter (const string & *diag*, bool *silent* = true)

Exception ctor, prefixes the string "FitsError: row offset or length incompatible with column declaration " before the specific message.

Parameters:

diag A specific diagnostic message, usually the column name

silent if true, print message whether FITS::verboseMode is set or not.

The documentation for this class was generated from the following files:

- Column.h
- Column.cxx

8.23 CCfits::Keyword Class Reference

Abstract base class implementing the common behavior of Keyword objects.

`#include <Keyword.h>`

Public Methods

- Keyword (const Keyword &right)
- Keyword (const string &keyname, ValueType keytype, HDU *p, const string &comment="")
- virtual ~Keyword ()
- Keyword& operator= (const Keyword &right)
- bool operator== (const Keyword &right) const
- bool operator!= (const Keyword &right) const
- virtual Keyword* clone () const=0
- virtual void write ()=0
- fitsfile* fitsPointer () const
- const std::string& name () const
- const std::string& comment () const

Protected Methods

- ValueType keytype () const
- void keytype (ValueType value)
- const HDU* parent () const

8.23.1 Detailed Description

Abstract base class implementing the common behavior of Keyword objects.

Keywords consists of a name, a value and a comment field. Concrete templated subclasses, KeyData<T>, have a data member that holds the value of keyword.

Typically, the mandatory keywords for a given HDU type are not stored as object of type Keyword, but as intrinsic data types. The Keyword hierarchy is used to store user-supplied information.

8.23.2 Constructor & Destructor Documentation

8.23.2.1 CCfits::Keyword::Keyword (const Keyword & *right*)

copy constructor.

8.23.2.2 CCfits::Keyword::Keyword (const string & *keyname*, ValueType *keytype*, HDU * *p*, const string & *comment* = "")

Keyword constructor.

This is the common behavior of Keywords of any type. Constructor is protected as the class is abstract.

8.23.2.3 CCfits::Keyword::~Keyword () [virtual]

virtual destructor.

8.23.3 Member Function Documentation

8.23.3.1 Keyword * CCfits::Keyword::clone () const [pure virtual]

virtual copy constructor.

8.23.3.2 const std::string & CCfits::Keyword::comment () const [inline]

return the comment field of the keyword.

8.23.3.3 fitsfile * CCfits::Keyword::fitsPointer () const

return a pointer to the FITS file containing the parent HDU.

8.23.3.4 void CCfits::Keyword::keytype (ValueType *value*) [inline, protected]

set keyword type.

8.23.3.5 ValueType CCfits::Keyword::keytype () const [inline, protected]

return the type of a keyword.

8.23.3.6 const std::string & CCfits::Keyword::name () const [inline]

return the name of a keyword

8.23.3.7 bool CCfits::Keyword::operator!= (const Keyword & *right*) const

inequality operator.

8.23.3.8 Keyword & CCfits::Keyword::operator= (const Keyword & *right*)

assignment operator.

8.23.3.9 bool CCfits::Keyword::operator== (const Keyword & *right*) const

equality operator.

8.23.3.10 const HDU * CCfits::Keyword::parent () const [inline, protected]

return a pointer to parent HDU.

8.23.3.11 void CCfits::Keyword::write () [pure virtual]

write operation.

The documentation for this class was generated from the following files:

- Keyword.h
- HDU.h
- Keyword.cxx
- KeywordT.h

8.24 CCfits::FITSUtil::MatchName Class Template Reference

predicate for classes that have a name attribute; match input string with instance name.

`#include <FITSUtil.h>`

8.24.1 Detailed Description

template<class T> class CCfits::FITSUtil::MatchName

predicate for classes that have a name attribute; match input string with instance name.

Usage: MatchName<NamedClass> Ex;

list<NamedClass> ListObject;

... ...

find_if(ListObject.begin(),ListObject().end(),bind2nd(Ex,"needle"));

Since most of the classes within CCfits are not implemented with lists, these functions are now of little direct use.

The documentation for this class was generated from the following file:

- FITSUtil.h

8.25 CCfits::FITSUtil::MatchNum Class Template Reference

predicate for classes that have an index attribute; match input index with instance value.

`#include <FITSUtil.h>`

8.25.1 Detailed Description

template<class T> class CCfits::FITSUtil::MatchNum

predicate for classes that have an index attribute; match input index with instance value.

Usage: MatchName<IndexedClass> Ex;

list<NamedClass> ListObject;

... ...

find_if(ListObject.begin(),ListObject().end(),bind2nd(Ex,5));

Since most of the classes within CCfits are implemented with std::maps rather than lists, these functions are now of little direct use.

The documentation for this class was generated from the following file:

- FITSUtil.h

8.26 CCfits::FITSUtil::MatchPtrName Class Template Reference

as for MatchName, only with the input class a pointer.

`#include <FITSUtil.h>`

8.26.1 Detailed Description

template<class T> class CCfits::FITSUtil::MatchPtrName

as for MatchName, only with the input class a pointer.

The documentation for this class was generated from the following file:

- FITSUtil.h

8.27 CCfits::FITSUtil::MatchPtrNum Class Template Reference

as for MatchNum, only with the input class a pointer.

`#include <FITSUtil.h>`

8.27.1 Detailed Description

template<class T> class CCfits::FITSUtil::MatchPtrNum

as for MatchNum, only with the input class a pointer.

The documentation for this class was generated from the following file:

- FITSUtil.h

8.28 CCfits::FITSUtil::MatchType Class Template Reference

function object that returns the FITS ValueType corresponding to an input intrinsic type.

`#include <FITSUtil.h>`

8.28.1 Detailed Description

template<typename T> class CCfits::FITSUtil::MatchType

function object that returns the FITS ValueType corresponding to an input intrinsic type.

This is particularly useful inside templated class instances where calls to cfitsio need to supply a value type. With this function one can extract the value type from the class type.

usage:

MatchType<T> type;

ValueType dataType = type();

Uses run-time type information (RTTI) methods.

The documentation for this class was generated from the following file:

- FITSUtil.h

8.29 CCfits::HDU::NoNullValue Class Reference

exception to be thrown on seek errors for keywords.

`#include <HDU.h>`

Inheritance diagram for CCfits::HDU::NoNullValue::

Public Methods

- NoNullValue (const string &diag, bool silent=true)

8.29.1 Detailed Description

exception to be thrown on seek errors for keywords.

8.29.2 Constructor & Destructor Documentation

8.29.2.1 CCfits::HDU::NoNullValue::NoNullValue (const string & *diag*, bool *silent* = true)

Exception ctor, prefixes the string ”Fits Error: No Null Pixel Value specified for Image ” before the specific message.

Parameters:

diag A specific diagnostic message, the name of the HDU if not the primary.

silent if true, print message whether FITS::verboseMode is set or not.

The documentation for this class was generated from the following files:

- HDU.h
- HDU.cxx

8.30 CCfits::Column::NoNullValue Class Reference

Exception thrown if a null value is specified without support from existing column header.

`#include <Column.h>`

Inheritance diagram for CCfits::Column::NoNullValue::

Public Methods

- NoNullValue (const string &diag, bool silent=true)

8.30.1 Detailed Description

Exception thrown if a null value is specified without support from existing column header.

This exception is analogous to the fact that cfitsio returns a non-zero status code if TNULLn doesn’t exist an a null value (convert all input data with the null value to the TNULLn keyword) is specified. It is only relevant for integer type data (see cfitsio manual for details).

8.30.2 Constructor & Destructor Documentation

8.30.2.1 CCfits::Column::NoNullValue::NoNullValue (const string & *diag*, bool *silent* = true)

Exception ctor, prefixes the string ”Fits Error: No null value specified for column: ” before the specific message.

Parameters:

diag A specific diagnostic message

silent if true, print message whether FITS::verboseMode is set or not.

The documentation for this class was generated from the following files:

- Column.h
- Column.cxx

8.31 CCfits::Table::NoSuchColumn Class Reference

Exception to be thrown on a failure to retrieve a column specified either by name or index number.

`#include <Table.h>`

Inheritance diagram for CCfits::Table::NoSuchColumn::

Public Methods

- NoSuchColumn (const string &name, bool silent=true)
- NoSuchColumn (int index, bool silent=true)

8.31.1 Detailed Description

Exception to be thrown on a failure to retrieve a column specified either by name or index number.

When a Table object is created, the header is read and a column object created for each column defined. Thus id this exception is thrown the column requested does not exist in the HDU (note that the column can easily exist and not contain any data since the user controls whether the column will be read when the FITS object is instantiated).

It is expected that the index number calls will be primarily internal. The underlying implementation makes lookup by name more efficient.

The exception has two variants, which take either an integer or a string as parameter. These are used according to the accessor that threw them, either by name or index.

8.31.2 Constructor & Destructor Documentation

8.31.2.1 CCfits::Table::NoSuchColumn::NoSuchColumn (const string & *name*, bool *silent* = true)

Exception ctor for exception thrown if the requested column (specified by name) is not present.

Message: Fits Error: cannot find Column named: *name* is printed.

Parameters:

name the requested column name

silent if true, print message whether FITS::verboseMode is set or not.

8.31.2.2 CCfits::Table::NoSuchColumn::NoSuchColumn (int *index*, bool *silent* = true)

Exception ctor for exception thrown if the requested column (specified by name) is not present.

Message: Fits Error: column not present - Column number *index* is printed.

Parameters:

index the requested column number

silent if true, print message whether FITS::verboseMode is set or not.

The documentation for this class was generated from the following files:

- Table.h
- Table.cxx

8.32 CCfits::FITS::NoSuchHDU Class Reference

exception thrown by HDU retrieval methods.

`#include <FITS.h>`

Inheritance diagram for CCfits::FITS::NoSuchHDU::

Public Methods

- NoSuchHDU (const string &diag, bool silent=true)

8.32.1 Detailed Description

exception thrown by HDU retrieval methods.

8.32.2 Constructor & Destructor Documentation

8.32.2.1 CCfits::FITS::NoSuchHDU::NoSuchHDU (const string & *diag*, bool *silent* = true)

Exception ctor, prefixes the string "FITS Error: Cannot read HDU in FITS file:" before the specific message.

Parameters:

diag A specific diagnostic message, usually the name of the extension whose read was attempted.

silent if true, print message whether FITS::verboseMode is set or not.

The documentation for this class was generated from the following files:

- FITS.h
- FITS.cxx

8.33 CCfits::HDU::NoSuchKeyword Class Reference

exception to be thrown on seek errors for keywords.

`#include <HDU.h>`

Inheritance diagram for CCfits::HDU::NoSuchKeyword::

Public Methods

- NoSuchKeyword (const string &diag, bool silent=true)

8.33.1 Detailed Description

exception to be thrown on seek errors for keywords.

8.33.2 Constructor & Destructor Documentation

8.33.2.1 CCfits::HDU::NoSuchKeyword::NoSuchKeyword (const string & *diag*, bool *silent* = true)

Exception ctor, prefixes the string "Fits Error: Keyword not found: " before the specific message.

Parameters:

diag A specific diagnostic message, usually the name of the keyword requested.

silent if true, print message whether FITS::verboseMode is set or not.

The documentation for this class was generated from the following files:

- HDU.h
- HDU.cxx

8.34 CCfits::FITS::OperationNotSupported Class Reference

thrown for unsupported operations, such as attempted to select rows from an image extension.

`#include <FITS.h>`

Inheritance diagram for CCfits::FITS::OperationNotSupported::

Public Methods

- OperationNotSupported (const string &msg, bool silent=true)

8.34.1 Detailed Description

thrown for unsupported operations, such as attempted to select rows from an image extension.

8.34.2 Constructor & Destructor Documentation

8.34.2.1 CCfits::FITS::OperationNotSupported::OperationNotSupported (const string & *msg*, bool *silent* = true)

Exception ctor, prefixes the string "FITS Error: Operation not supported:" before the specific message.

Parameters:

msg A specific diagnostic message.

silent if true, print message whether FITS::verboseMode is set or not.

The documentation for this class was generated from the following files:

- FITS.h
- FITS.cxx

8.35 CCfits::PHDU Class Reference

class representing the primary HDU for a FITS file.

`#include <PHDU.h>`

Inheritance diagram for CCfits::PHDU::

Public Methods

- virtual PHDU* clone (FITSBase *p) const=0
- virtual void zero (double value)
- virtual double zero () const
- virtual double scale () const
- template<typename S> void write (const std::vector< long > &first, long nElements, const std::valarray< S > &data, S *nullValue)
- template<typename S> void write (long first, long nElements, const std::valarray< S > &data, S *nullValue)
- template<typename S> void write (const std::vector< long > &first, long nElements, const std::valarray< S > &data)
- template<typename S> void write (long first, long nElements, const std::valarray< S > &data)
- template<typename S> void write (const std::vector< long > &firstVertex, const std::vector< long > &lastVertex, const std::valarray< S > &data)
- template<typename S> void read (std::valarray< S > &image, long first, long nElements)
- template<typename S> void read (std::valarray< S > &image, long first, long nElements, S *nullValue)
- template<typename S> void read (std::valarray< S > &image, const std::vector< long > &first, long nElements)
- template<typename S> void read (std::valarray< S > &image, const std::vector< long > &first, long nElements, S *nullValue)
- template<typename S> void read (std::valarray< S > &image, const std::vector< long > &firstVertex, const std::vector< long > &lastVertex, const std::vector< long > &stride)
- template<typename S> void read (std::valarray< S > &image, const std::vector< long > &firstVertex, const std::vector< long > &lastVertex, const std::vector< long > &stride, S *nullValue)

Protected Methods

- PHDU (const PHDU &right)
- PHDU (FITSBase *p, int bpix, int naxis, const std::vector< long > &axes)
- PHDU (FITSBase *p=0)

8.35.1 Detailed Description

class representing the primary HDU for a FITS file.

A PHDU object is automatically instantiated and added to a FITS object when a FITS file is accessed in any way. If a new file is created without specifying the data type for the header, CCfits assumes that the file is to be used for table extensions and creates a dummy header. PHDU instances are *only* created by FITS ctors. In the first release of CCfits, the Primary cannot be changed once declared.

PHDU and ExtHDU provide the same interface to writing images: multiple overloads of the templated PHDU::read and PHDU::write operations provide for (a) writing image data specified in a number of ways [C-array, std::vector, std::valarray] and with input location specified by initial pixel, by n-tuple, and by rectangular subset [generalized slice]; (b) reading image data specified similarly to the write options into a std::valarray.

Todo:

Implement functions that allow replacement of the primary image

8.35.2 Constructor & Destructor Documentation

8.35.2.1 CCfits::PHDU::PHDU (const PHDU & *right*) [protected]

copy constructor.

required for cloning primary HDUs when copying FITS files.

8.35.2.2 CCfits::PHDU::PHDU (FITSBase * *p*, int *bpix*, int *naxis*, const std::vector< long > & *axes*) [protected]

Writing Primary HDU constructor, called by PrimaryHDU<T> class.

Constructor used for creating new PHDU (i.e. for writing data to FITS). also doubles as default constructor since all arguments have default values, which are passed to the HDU constructor

8.35.2.3 CCfits::PHDU::PHDU (FITSBase * *p* = 0) [protected]

Reading Primary HDU constructor.

Constructor used when reading the primary HDU from an existing file. Does nothing except initialize, with the real work done by the subclass PrimaryHDU<T>.

8.35.3 Member Function Documentation

8.35.3.1 PHDU * CCfits::PHDU::clone (FITSBase * *p*) const [pure virtual]

virtual copy constructor, to be implemented in subclasses.

Reimplemented from CCfits::HDU.

8.35.3.2 template<typename S> void CCfits::PHDU::read (std::valarray< S > & *image*, const std::vector< long > & *firstVertex*, const std::vector< long > & *lastVertex*, const std::vector< long > & *stride*, S * *nullValue*)

read an image subset into valarray image, processing null values.

The image subset is defined by two vertices and a stride indicating the 'denseness' of the values to be picked in each dimension (a stride = (1,1,1,...) means picking every pixel in every dimension, whereas stride = (2,2,2,...) means picking every other value in each dimension.

8.35.3.3 template<typename S> void CCfits::PHDU::read (std::valarray< S > & *image*, const std::vector< long > & *firstVertex*, const std::vector< long > & *lastVertex*, const std::vector< long > & *stride*)

read an image subset.

8.35.3.4 template<typename S> void CCfits::PHDU::read (std::valarray< S > & *image*, const std::vector< long > & *first*, long *nElements*, S * *nullValue*)

read part of an image array, processing null values.

As above except for

Parameters:

first a vector<long> representing an n-tuple giving the coordinates in the image of the first pixel.

8.35.3.5 template<typename S> void CCfits::PHDU::read (std::valarray< S > & *image*, const std::vector< long > & *first*, long *nElements*)

read an image section starting at a location specified by an n-tuple.

8.35.3.6 template<typename S> void CCfits::PHDU::read (std::valarray< S > & *image*, long *first*, long *nElements*, S * *nullValue*)

read part of an image array, processing null values.

Implicit data conversion is supported (i.e. user does not need to know the type of the data stored. A WrongExtensionType extension is thrown if *this is not an image.

Parameters:

image The receiving container, a std::valarray reference

first The first pixel from the array to read [a long value]

nElements The number of values to read

nullValue A pointer containing the value in the table to be considered as undefined. See cfitsio for details

8.35.3.7 template<typename S> void CCfits::PHDU::read (std::valarray< S > & *image*, long *first*, long *nElements*)

read an image section starting at a specified pixel.

8.35.3.8 double CCfits::PHDU::scale () const [virtual]

return the BSCALE keyword value.

Reimplemented from CCfits::HDU.

8.35.3.9 template<typename S> void CCfits::PHDU::write (const std::vector< long > & *firstVertex*, const std::vector< long > & *lastVertex*, const std::valarray< S > & *data*)

write a subset (generalize slice) of data to the image.

A generalized slice/subset is a subset of the image (e.g. one plane of a data cube of size <= the dimension of the cube). It is specified by two opposite vertices. The equivalent cfitsio call does not support undefined data processing so there is no version that allows a null value to be specified.

Parameters:

firstVertex the coordinates specifying lower and upper vertices of the n-dimensional slice

lastVertex

data The data to be written

8.35.3.10 template<typename S> void CCfits::PHDU::write (long *first*, long *nElements*, const std::valarray< S > & *data*)

write array starting from specified pixel number, without undefined data processing.

8.35.3.11 template<typename S> void CCfits::PHDU::write (const std::vector< long > & *first*, long *nElements*, const std::valarray< S > & *data*)

write array starting from specified n-tuple, without undefined data processing.

8.35.3.12 template<typename S> void CCfits::PHDU::write (long *first*, long *nElements*, const std::valarray< S > & *data*, S * *nullValue*)

write array to image starting with a specified pixel and allowing undefined data to be processed.

parameters after the first are as for version with n-tuple specifying first element. these two version are equivalent, except that it is possible for the first pixel number to exceed the range of 32-bit integers, which is how long datatype is commonly implemented.

8.35.3.13 template<typename S> void CCfits::PHDU::write (const std::vector< long > & *first*, long *nElements*, const std::valarray< S > & *data*, S * *nullValue*)

Write a set of pixels to an image extension with the first pixel specified by an n-tuple, processing undefined data.

All the overloaded versions of PHDU::write perform operations on *this if it is an image and throw a WrongExtensionType exception if not. Where appropriate, alternate versions allow undefined data to be processed

Parameters:

first an n-tuple of dimension equal to the image dimension specifying the first pixel in the range to be written

nElements number of pixels to be written

data array of data to be written

pointer to null value (data with this value written as undefined; needs the BLANK keyword to have been specified).

8.35.3.14 double CCfits::PHDU::zero () const [virtual]

return the BZERO keyword value.

Reimplemented from CCfits::HDU.

8.35.3.15 void CCfits::PHDU::zero (double *value*) [virtual]

Private: called by ctor.

Reimplemented from CCfits::HDU.

The documentation for this class was generated from the following files:

- PHDU.h
- PHDU.cxx
- PHDUT.h

8.36 CCfits::Column::RangeError Class Reference

exception to be thrown for inputs that cause range errors in column read operations.

`#include <Column.h>`

Inheritance diagram for CCfits::Column::RangeError::

Public Methods

- RangeError (const string &msg, bool silent=true)

8.36.1 Detailed Description

exception to be thrown for inputs that cause range errors in column read operations.

Range errors here mean (last < first) in a request to read a range of rows.

8.36.2 Constructor & Destructor Documentation

8.36.2.1 CCfits::Column::RangeError::RangeError (const string & *msg*, bool *silent* = true)

Exception ctor, prefixes the string "FitsError: Range error in operation " before the specific message.

Parameters:

msg A specific diagnostic message

silent if true, print message whether FITS::verboseMode is set or not.

The documentation for this class was generated from the following files:

- Column.h
- Column.cxx

8.37 CCfits::Table Class Reference

`#include <Table.h>`

Inheritance diagram for CCfits::Table::

Public Methods

- Table (const Table &right)
- virtual ~Table ()
- const std::map<string,Column*>& column () const
- virtual Column& column (const string &colName) const
- virtual Column& column (int colIndex) const
- virtual long rows () const
- void updateRows ()
- void rows (long numRows)
- virtual void deleteColumn (const string &name)
- void insertRows (long first, long number=1)
- void deleteRows (long first, long number=1)
- void deleteRows (const std::vector< long > &rowList)
- virtual std::map<string, Column*>& column ()

Protected Methods

- Table (FITSBase *p, HduType xtype, const string &hduName, int rows, const std::vector< string > &columnName, const std::vector< string > &columnFmt, const std::vector< string > &columnUnit=std::vector< string >(), int version=1)
- Table (FITSBase *p, HduType xtype, const string &hduName=string(""), int version=1)
- Table (FITSBase *p, HduType xtype, int number)
- void init (bool readFlag=false, const std::vector< string > &keys=std::vector< string >())
- virtual void column (const string &colname, Column *value)
- int numCols () const
- void numCols (int value)

8.37.1 Detailed Description

Table is the abstract common interface to Binary and Ascii Table HDUs.

Table is a subclass of ExtHDU that contains an associative array of Column objects. It implements methods for reading and writing columns

8.37.2 Constructor & Destructor Documentation

8.37.2.1 CCfits::Table::Table (const Table & *right*)

copy constructor.

8.37.2.2 CCfits::Table::~Table () [virtual]

destructor.

8.37.2.3 CCfits::Table::Table (FITSBase * *p*, HduType *xtype*, const string & *hduName*, int *rows*, const std::vector< string > & *columnName*, const std::vector< string > & *columnFmt*, const std::vector< string > & *columnUnit* = std::vector<string>(), int *version* = 1) [protected]

Constructor to be used for creating new HDUs.

Parameters:

p The FITS file in which to place the new HDU

rows The number of rows in the new HDU (the value of the NAXIS2 keyword).

columnType a vector of types for the columns.

columnName a vector of names for the columns.

columnFmt the format strings for the columns

columnUnit the units for the columns.

version a version number

8.37.2.4 CCfits::Table::Table (FITSBase * *p*, HduType *xtype*, const string & *hduName* = string(""), int *version* = 1) [protected]

Constructor to be called by operations that read Table specified by hduName and version.

Parameters:

xtype The type of the Table (AsciiTbl or BinaryTbl)

8.37.2.5 CCfits::Table::Table (FITSBase * *p*, HduType *xtype*, int *number*) [protected]

Table constructor for getting Tables by number.

Necessary since EXTNAME is a reserved not required keyword, and users may thus read FITS files without an extension name. Since an HDU is completely specified by extension number, this is part of the public interface.

8.37.3 Member Function Documentation

8.37.3.1 void CCfits::Table::column (const string & *colname*, Column * *value*) [protected, virtual]

set the column with name colname to the input value.

Reimplemented from CCfits::ExtHDU.

8.37.3.2 std::map< string, Column * > & CCfits::Table::column () [inline, virtual]

return a reference to the array containing the columns.

To be used in the implementation of subclasses.

8.37.3.3 Column & CCfits::Table::column (int *colIndex*) const [virtual]

return a reference to the column identified by colIndex.

Throws NoSuchColumn if the index is out of range -index must satisfy (1 <= index <= numCols()).

N.B. the column number is assigned as 1-based, as in FORTRAN rather than 0-based as in C.

Exceptions:

Table::NoSuchColumn passes colIndex to the diagnostic message printed when the exception is thrown

Reimplemented from CCfits::ExtHDU.

8.37.3.4 Column & CCfits::Table::column (const string & *colName*) const [virtual]

return a reference to the column of name colName.

Exceptions:

Table::NoSuchColumn passes colName to the diagnostic message printed when the exception is thrown

Reimplemented from CCfits::ExtHDU.

8.37.3.5 const std::map< string, Column * > & CCfits::Table::column () const [inline]

return a reference to the array containing the columns.

This public version might be used to query the size of the column container in a routine that manipulates column table data.

8.37.3.6 void CCfits::Table::deleteColumn (const string & *name*) [virtual]

delete a column in a Table extension by name.

Parameters:

columnName The name of the column to be deleted.

Exceptions:
WrongExtensionType if extension is an image.

Reimplemented from CCfits::ExtHDU.

8.37.3.7 void CCfits::Table::deleteRows (const std::vector< long > & *rowList*)

delete a set of rows in the table specified by an input array.

Parameters:
rowlist The vector of row numbers to be deleted.

Exceptions:
FitsError thrown if the underlying cfitsio call fails to return without error.

8.37.3.8 void CCfits::Table::deleteRows (long *first*, long *number* = 1)

delete a range of rows in a table.

In both this and the overloaded version which allows a selection of rows to be deleted, the cfitsio library is called first to perform the operation on the disk file, and then the FITS object is updated.

Parameters:
first the start row of the range

number the number of rows to delete; defaults to 1.

Exceptions:
FitsError thrown if the cfitsio call fails to return without error.

8.37.3.9 void CCfits::Table::init (bool *readFlag* = false, const std::vector< string > & *keys* = std::vector<string>()) [protected]

"Late Constructor." wrap-up of calls needed to construct a table. Reads header information and sets up the array of column objects in the table.

Protected function, provided to allow the implementation of extensions of the library.

8.37.3.10 void CCfits::Table::insertRows (long *first*, long *number* = 1)

insert empty rows into the table.

Parameters:
first the start row of the range

number the number of rows to insert.

Exceptions:

FitsError thrown if the underlying cfitsio call fails to return without error.

8.37.3.11 void CCfits::Table::numCols (int *value*) [inline, protected]

set the number of Columns in the Table.

8.37.3.12 int CCfits::Table::numCols () const [inline, protected]

return the number of Columns in the Table (the TFIELDS keyword).

8.37.3.13 void CCfits::Table::rows (long *numRows*) [inline]

set the number of rows in the Table.

8.37.3.14 long CCfits::Table::rows () const [inline, virtual]

return the number of rows in the table (NAXIS2).

Reimplemented from CCfits::ExtHDU.

8.37.3.15 void CCfits::Table::updateRows ()

update the number of rows in the table.

Called to force the Table to reset its internal ”rows” attribute. public, but is called when needed internally.

The documentation for this class was generated from the following files:

- Table.h
- Column.h
- Table.cxx

8.38 CCfits::FITSUtil::MatchType::UnrecognizedType Class Template Reference

exception thrown by MatchType if it encounters data type incompatible with cfitsio.

`#include <FITSUtil.h>`

8.38.1 Detailed Description

template<typename T> class CCfits::FITSUtil::MatchType::UnrecognizedType

exception thrown by MatchType if it encounters data type incompatible with cfitsio.

The documentation for this class was generated from the following file:

- FITSUtil.h

8.39 CCfits::Column::WrongColumnType Class Reference

Exception thrown on attempting to access a scalar column as vector data.

`#include <Column.h>`

Inheritance diagram for CCfits::Column::WrongColumnType::

Public Methods

- WrongColumnType (const string &diag, bool silent=true)

8.39.1 Detailed Description

Exception thrown on attempting to access a scalar column as vector data.

This exception will be thrown if the user tries to call a read/write operation with a signature appropriate for a vector column on a scalar column, or vice versa. For example in the case of write operations, the vector versions require the specification of (a) a number of rows to write over, (b) a vector of lengths to write to each row or (c) a subset specification. The scalar versions only require a number of rows if the input array is a plain C-array, otherwise the range to be written is the size of the input vector.

8.39.2 Constructor & Destructor Documentation

8.39.2.1 CCfits::Column::WrongColumnType::WrongColumnType (const string & *diag*, bool *silent* = true)

Exception ctor, prefixes the string "FitsError: Attempt to return scalar data from vector column, or vice versa - Column: " before the specific message.

Parameters:

diag A specific diagnostic message, usually the column name.

silent if true, print message whether FITS::verboseMode is set or not.

The documentation for this class was generated from the following files:

- Column.h
- Column.cxx

8.40 CCfits::ExtHDU::WrongExtensionType Class Reference

Exception to be thrown on unmatched extension types.

`#include <ExtHDU.h>`

Inheritance diagram for CCfits::ExtHDU::WrongExtensionType::

Public Methods

- WrongExtensionType (const string &msg, bool silent=true)

8.40.1 Detailed Description

Exception to be thrown on unmatched extension types.

This exception is to be thrown if the user requested a particular extension and it does not correspond to the expected type.

8.40.2 Constructor & Destructor Documentation

8.40.2.1 CCfits::ExtHDU::WrongExtensionType::WrongExtensionType (const string & *msg*, bool *silent* = true)

Exception ctor, prefixes the string "Fits Error: wrong extension type" before the specific message.

Parameters:

msg A specific diagnostic message

silent if true, print message whether FITS::verboseMode is set or not.

The documentation for this class was generated from the following files:

- ExtHDU.h
- ExtHDU.cxx

Index

~AsciiTable
    CCfits::AsciiTable, 40
~BinTable
    CCfits::BinTable, 46
~Column
    CCfits::Column, 53
~ExtHDU
    CCfits::ExtHDU, 67
~FITS
    CCfits::FITS, 81
~HDU
    CCfits::HDU, 92
~Keyword
    CCfits::Keyword, 106
~Table
    CCfits::Table, 126
~auto_array_ptr
    CCfits::FITSUtil::auto_array_ptr, 42

addColumn
    CCfits::AsciiTable, 40
    CCfits::BinTable, 46
    CCfits::ExtHDU, 68
addImage
    CCfits::FITS, 81
addKey
    CCfits::HDU, 93
addNullValue
    CCfits::Column, 54
addTable
    CCfits::FITS, 81
AsciiTable
    CCfits::AsciiTable, 38, 39
auto_array_ptr
    CCfits::FITSUtil::auto_array_ptr, 42
axes
    CCfits::HDU, 93
axis
    CCfits::HDU, 93

BinTable
    CCfits::BinTable, 45, 46
bitpix
    CCfits::HDU, 93

CantCreate
    CCfits::FITS::CantCreate, 49
CantOpen
    CCfits::FITS::CantOpen, 50
CCfits::AsciiTable, 37
    ~AsciiTable, 40
    addColumn, 40
    AsciiTable, 38, 39
    clone, 40
    readData, 40
CCfits::BinTable, 44
    ~BinTable, 46
    addColumn, 46
    BinTable, 45, 46
    clone, 47
    readData, 47
CCfits::Column, 50
    ~Column, 53
    addNullValue, 54
    Column, 53, 54
    comment, 54
    dimen, 54
    display, 54
    fitsPointer, 54
    format, 55
    index, 55
    isRead, 55
    makeHDUCurrent, 55
    name, 55
    put, 55
    read, 55–57
    readArrays, 57
    readData, 57
    repeat, 58
    rows, 58
    scale, 58
    setDimen, 58
    setDisplay, 58
    type, 58

unit, 58
varLength, 59
width, 59
write, 59–62
writeArrays, 63
zero, 63
CCfits::Column::InsufficientElements, 98
InsufficientElements, 98
CCfits::Column::InvalidDataType, 99
InvalidDataType, 99
CCfits::Column::InvalidNumberOfRows, 102
InvalidNumberOfRows, 102
CCfits::Column::InvalidRowNumber, 103
InvalidRowNumber, 103
CCfits::Column::InvalidRowParameter, 104
InvalidRowParameter, 105
CCfits::Column::NoNullValue, 112
NoNullValue, 112
CCfits::Column::RangeError, 123
RangeError, 124
CCfits::Column::WrongColumnType, 131
WrongColumnType, 131
CCfits::ExtHDU, 65
∼ExtHDU, 67
addColumn, 68
clone, 69
column, 69
deleteColumn, 69
ExtHDU, 67, 68
gcount, 70
makeThisCurrent, 70
name, 70
pcount, 70
read, 71, 72
readData, 72
readHduName, 72
rows, 72
version, 73
write, 73, 74
xtension, 74
CCfits::ExtHDU::WrongExtensionType, 132
WrongExtensionType, 132
CCfits::FITS, 75
∼FITS, 81
addImage, 81
addTable, 81
clearErrors, 82
clone, 82
copy, 82
currentExtension, 82
currentExtensionName, 82
deleteExtension, 82, 83
destroy, 83
extension, 83, 84
filter, 84
FITS, 77–80
flush, 84
name, 84
pHDU, 85
read, 85, 86
resetPosition, 86
setVerboseMode, 86
verboseMode, 86
CCfits::FITS::CantCreate, 48
CantCreate, 49
CCfits::FITS::CantOpen, 49
CantOpen, 50
CCfits::FITS::NoSuchHDU, 115
NoSuchHDU, 115
CCfits::FITS::OperationNotSupported, 117
OperationNotSupported, 117
CCfits::FitsError, 87
FitsError, 87
CCfits::FitsException, 88
FitsException, 89
CCfits::FitsFatal, 89
FitsFatal, 90
CCfits::FITSUtil::auto_array_ptr, 41
∼auto_array_ptr, 42
auto_array_ptr, 42
get, 42
operator ∗, 42
operator=, 43
operator[], 43

release, 43
remove, 43
reset, 43
CCfits::FITSUtil::CAarray, 47
operator(), 48
CCfits::FITSUtil::CVAarray, 63
operator(), 64
CCfits::FITSUtil::CVarray, 64
operator(), 65
CCfits::FITSUtil::MatchName, 108
CCfits::FITSUtil::MatchNum, 109
CCfits::FITSUtil::MatchPtrName, 109
CCfits::FITSUtil::MatchPtrNum, 110
CCfits::FITSUtil::MatchType, 110
CCfits::FITSUtil::MatchType::UnrecognizedType, 130
CCfits::HDU, 90
~HDU, 92
addKey, 93
axes, 93
axis, 93
bitpix, 93
clone, 94
comment, 94
deleteKey, 94
fitsPointer, 94
getComments, 94
getHistory, 94
HDU, 92
history, 94
index, 95
keyWord, 95
makeThisCurrent, 95
naxes, 96
operator!=, 96
operator==, 96
parent, 96
readAllKeys, 96
readKey, 96
readKeys, 96
scale, 97
setKeyWord, 97
writeComment, 97
writeDate, 97
writeHistory, 97
zero, 97
CCfits::HDU::InvalidExtensionType, 100
InvalidExtensionType, 100
CCfits::HDU::InvalidImageDataType, 101
InvalidImageDataType, 101
CCfits::HDU::NoNullValue, 111
NoNullValue, 111
CCfits::HDU::NoSuchKeyword, 116
NoSuchKeyword, 116
CCfits::Keyword, 105
~Keyword, 106
clone, 106
comment, 107
fitsPointer, 107
keytype, 107
Keyword, 106
name, 107
operator!=, 107
operator=, 107
operator==, 107
parent, 108
write, 108
CCfits::PHDU, 118
clone, 120
PHDU, 119
read, 120, 121
scale, 121
write, 121, 122
zero, 123
CCfits::Table, 124
~Table, 126
column, 127, 128
deleteColumn, 128
deleteRows, 128, 129
init, 129
insertRows, 129
numCols, 129, 130
rows, 130
Table, 126, 127
updateRows, 130
CCfits::Table::NoSuchColumn, 113
NoSuchColumn, 114
clearErrors
CCfits::FITS, 82

clone
CCfits::AsciiTable, 40
CCfits::BinTable, 47
CCfits::ExtHDU, 69
CCfits::FITS, 82
CCfits::HDU, 94
CCfits::Keyword, 106
CCfits::PHDU, 120
Column
CCfits::Column, 53, 54
column
CCfits::ExtHDU, 69
CCfits::Table, 127, 128
comment
CCfits::Column, 54
CCfits::HDU, 94
CCfits::Keyword, 107
copy
CCfits::FITS, 82
currentExtension
CCfits::FITS, 82
currentExtensionName
CCfits::FITS, 82

deleteColumn
CCfits::ExtHDU, 69
CCfits::Table, 128
deleteExtension
CCfits::FITS, 82, 83
deleteKey
CCfits::HDU, 94
deleteRows
CCfits::Table, 128, 129
destroy
CCfits::FITS, 83
dimen
CCfits::Column, 54
display
CCfits::Column, 54

extension
CCfits::FITS, 83, 84
ExtHDU
CCfits::ExtHDU, 67, 68

filter
CCfits::FITS, 84
FITS
CCfits::FITS, 77–80
FITS Exceptions, 35
FitsError
CCfits::FitsError, 87
FitsException
CCfits::FitsException, 89
FitsFatal
CCfits::FitsFatal, 90
fitsPointer
CCfits::Column, 54
CCfits::HDU, 94
CCfits::Keyword, 107
FITSUtil, 37
flush
CCfits::FITS, 84
format
CCfits::Column, 55

gcount
CCfits::ExtHDU, 70
get
CCfits::FITSUtil::auto_array_ptr, 42
getComments
CCfits::HDU, 94
getHistory
CCfits::HDU, 94

HDU
CCfits::HDU, 92
history
CCfits::HDU, 94

index
CCfits::Column, 55
CCfits::HDU, 95
init
CCfits::Table, 129
insertRows
CCfits::Table, 129
InsufficientElements
CCfits::Column::Insufficient-Elements, 98
InvalidDataType

CCfits::Column::InvalidDataType, 99
InvalidExtensionType
CCfits::HDU::InvalidExtensionType, 100
InvalidImageDataType
CCfits::HDU::InvalidImageDataType, 101
InvalidNumberOfRows
CCfits::Column::InvalidNumberOfRows, 102
InvalidRowNumber
CCfits::Column::InvalidRowNumber, 103
InvalidRowParameter
CCfits::Column::InvalidRowParameter, 105
isRead
CCfits::Column, 55

keytype
CCfits::Keyword, 107
Keyword
CCfits::Keyword, 106
keyWord
CCfits::HDU, 95

makeHDUCurrent
CCfits::Column, 55
makeThisCurrent
CCfits::ExtHDU, 70
CCfits::HDU, 95

name
CCfits::Column, 55
CCfits::ExtHDU, 70
CCfits::FITS, 84
CCfits::Keyword, 107
naxes
CCfits::HDU, 96
NoNullValue
CCfits::Column::NoNullValue, 112
CCfits::HDU::NoNullValue, 111
NoSuchColumn
CCfits::Table::NoSuchColumn, 114
NoSuchHDU
CCfits::FITS::NoSuchHDU, 115
NoSuchKeyword
CCfits::HDU::NoSuchKeyword, 116
numCols
CCfits::Table, 129, 130

OperationNotSupported
CCfits::FITS::OperationNotSupported, 117
operator ∗
CCfits::FITSUtil::auto_array_ptr, 42
operator!=
CCfits::HDU, 96
CCfits::Keyword, 107
operator()
CCfits::FITSUtil::CAarray, 48
CCfits::FITSUtil::CVAarray, 64
CCfits::FITSUtil::CVarray, 65
operator=
CCfits::FITSUtil::auto_array_ptr, 43
CCfits::Keyword, 107
operator==
CCfits::HDU, 96
CCfits::Keyword, 107
operator[]
CCfits::FITSUtil::auto_array_ptr, 43

parent
CCfits::HDU, 96
CCfits::Keyword, 108
pcount
CCfits::ExtHDU, 70
PHDU
CCfits::PHDU, 119
pHDU
CCfits::FITS, 85
put
CCfits::Column, 55

RangeError
CCfits::Column::RangeError, 124

read
CCfits::Column, 55–57
CCfits::ExtHDU, 71, 72
CCfits::FITS, 85, 86
CCfits::PHDU, 120, 121
readAllKeys
CCfits::HDU, 96
readArrays
CCfits::Column, 57
readData
CCfits::AsciiTable, 40
CCfits::BinTable, 47
CCfits::Column, 57
CCfits::ExtHDU, 72
readHduName
CCfits::ExtHDU, 72
readKey
CCfits::HDU, 96
readKeys
CCfits::HDU, 96
release
CCfits::FITSUtil::auto_array_ptr, 43
remove
CCfits::FITSUtil::auto_array_ptr, 43
repeat
CCfits::Column, 58
reset
CCfits::FITSUtil::auto_array_ptr, 43
resetPosition
CCfits::FITS, 86
rows
CCfits::Column, 58
CCfits::ExtHDU, 72
CCfits::Table, 130

scale
CCfits::Column, 58
CCfits::HDU, 97
CCfits::PHDU, 121
setDimen
CCfits::Column, 58
setDisplay
CCfits::Column, 58
setKeyWord
CCfits::HDU, 97
setVerboseMode
CCfits::FITS, 86

Table
CCfits::Table, 126, 127
type
CCfits::Column, 58

unit
CCfits::Column, 58
updateRows
CCfits::Table, 130

varLength
CCfits::Column, 59
verboseMode
CCfits::FITS, 86
version
CCfits::ExtHDU, 73

width
CCfits::Column, 59
write
CCfits::Column, 59–62
CCfits::ExtHDU, 73, 74
CCfits::Keyword, 108
CCfits::PHDU, 121, 122
writeArrays
CCfits::Column, 63
writeComment
CCfits::HDU, 97
writeDate
CCfits::HDU, 97
writeHistory
CCfits::HDU, 97
WrongColumnType
CCfits::Column::WrongColumnType, 131
WrongExtensionType
CCfits::ExtHDU::WrongExtensionType, 132

xtension
CCfits::ExtHDU, 74

zero
CCfits::Column, 63
CCfits::HDU, 97
CCfits::PHDU, 123